Bi-SSFormer: An Ultralightweight Binary Spectral–Spatial Transformer for Hyperspectral Image Classification

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***Abstract*—With the advancement of deep learning techniques, deep neural networks have progressively supplanted traditional machine learning methods for hyperspectral image (HSI) classification, demonstrating the superior performance across diverse datasets. Contemporary deep learning approaches for HSI classification fall primarily into three categories: convolutional neural networks (CNNs), Transformer-based models, and CNN–Transformer hybrid architectures. Driven by real-world application demands, the development of lightweight CNN–Transformer models has emerged as a significant research focus, aiming to optimize the balance between model complexity and performance. Nevertheless, these methods predominantly represent model weights using floating-point values, necessitating computationally intensive floating-point multiplication operations within convolutional and linear layers. To address this limitation, we propose a novel ultralightweight binary spectral–spatial transformer featuring binarized weights named Bi-SSFormer. This approach achieves state-of-the-art (SOTA) performance while minimizing computational and memory requirements. Specifically, we first construct a lightweight CNN–Transformer framework built upon specialized spectral–spatial blocks (SS-Blocks) which are composed of spectral and spatial interaction modules (SpeIM and SpaIM). SpeIM employs intragroup convolution and cross-group convolution to facilitate cross-spectral feature interaction, while SpaIM integrates linear-complexity local and global attention mechanisms for spatial feature interaction and fusion. Furthermore, we introduce an information-enhanced weight binarization (InWB) method, which attains a 32× reduction in weight storage and implements the multiplication-free convolutional and linear layers. Comprehensive experiments on four benchmark HSI datasets demonstrate that our proposed method achieves SOTA classification results, such as an overall accuracy (OA) of 96.55% on Indian Pines dataset while requiring less than 0.02 MB of memory occupation and 4.91M FLOPs. Compared with other SOTA lightweight models, our method strikes an optimal balance between the model performance and complexity.**

***Index Terms*—Convolutional neural network (CNN)–Transformer hybrid models, hyperspectral image (HSI) classification, lightweight networks, remote sensing, Transformers, weight binarization.**

I. Introduction

HYPERSPECTRAL imaging detects and collects electromagnetic spectra from the visible to near-infrared wavelength ranges during Earth observation [1], [2]. Benefiting from rich information in hundreds of narrow contiguous spectral bands [3], hyperspectral images (HSIs) have been successfully applied in diverse real-world applications including agricultural production [4], [5], [6], [7], mineral exploration [8], [9], [10], [11], environmental monitoring [12], [13], [14], [15], and underwater target detection [16], [17], [18], [19]. To support these high-level tasks, HSI classification has attracted increasing attention in the remote sensing community. This task aims to assign specific class labels to each pixel in HSIs [20], [21].

With the rapid development of deep neural networks [22], [23], [24], [25], [26] and their successful applications in remote sensing [27], [28], [29], HSI classification performance has also improved remarkably, transitioning from machine learning-based (ML-based) methods [30], [31], [32] to deep learning-based (DL-based) approaches [33]. Consequently, effective deep learning architecture design has become a prominent research focus in recent years, with models broadly categorized as convolutional neural networks (CNNs), Transformer-based models, and CNN–Transformer hybrid architectures. CNN-based models typically stack consecutive 2-D or 3-D convolutional layers to hierarchically extract spectral–spatial features [34], [35], achieving significant performance gains over ML-based approaches. Nevertheless, these models suffer from limited representational capacity due to the constrained receptive fields of the convolutional operations, leading to suboptimal performance on certain HSI classification tasks. To address this limitation, Transformer-based methods leverage multihead self-attention (MHSA) mechanisms to model long-range depen-

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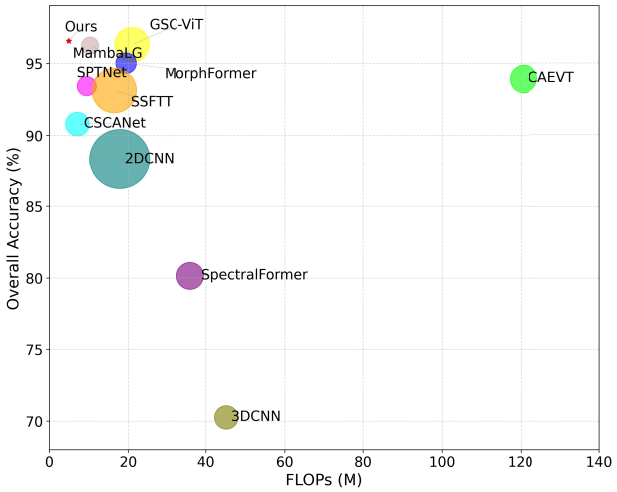


Fig. 1. OA against FLOPs of the proposed model and other advanced approaches on Indian Pines dataset. The circle areas are proportional to the model size. Our model achieves the highest OA with the fewest FLOPs and memory requirements.

dencies across spectral signatures [36], [37], capturing global contextual information to improve classification. Despite this advantage, Transformers' inherent lack of local feature extraction capabilities may compromise prediction accuracy. To overcome these complementary challenges, CNN–Transformer hybrid models integrate both architectures' strengths: convolutional layers extract local features while Transformer layers model global dependencies. For instance, Zhang et al. [38] employed convolutional autoencoders to capture nonlinear local features from spectral bands, subsequently constructing powerful vision Transformers for global feature extraction. This hybrid approach demonstrates superior performance, particularly under limited sample conditions.

However, considering the large number of spectral channels in HSI input data, conventional convolutional and Transformer layers introduce high computational and memory complexity. This impedes efficient HSI classification deployment on resource-constrained hardware. Consequently, emerging research focuses on developing lightweight CNN–Transformer hybrid models for effective yet efficient classification. Zhao et al. [39] proposed the groupwise separable convolutional Vision Transformer (GSC-ViT) to reduce memory and computation budgets while maintaining state-of-the-art (SOTA) performance on benchmark HSI datasets. Zhang et al. [40] employed the principal component analysis (PCA) for dimensionality reduction to manage computational costs, proposing a cascaded spatial cross-attention network (CSCANet) for efficient local–global integration. Although these methods reduce model complexity to some extent, parameters remain represented as 32-bit floating-point values, and substantial floating-point multiplication operations persist in convolutional and Transformer layers. Thus, significant compression opportunities remain for current lightweight HSI classification models.

In this article, to advance efficient network design, as demonstrated in Fig. 1, we propose an ultralightweight binary spectral–spatial transformer featuring binarized weights (Bi-SSFormer), which strikes the best balance between model performance and complexity. This approach significantly reduces memory requirements and eliminates resource-intensive floating-point multiplications. Specifically, we first construct an efficient CNN–Transformer hybrid model built on the spectral–spatial blocks (SS-Blocks) which comprise spectral interaction modules (SpeIMs) and spatial interaction modules (SpaIMs). The SpeIM integrates intragroup and cross-group convolutional layers to enhance intergroup spectral feature interactions. The SpaIM incorporates window and grid attention mechanisms for local and global feature extraction, respectively, followed by a fusion module integrating spatial information. Finally, we introduce an information-enhanced weight binarization (InWB) method, constraining model parameters to binary values (+1/−1). This reduces weight storage to 1-bit while replacing multiplication–addition operations in linear and convolutional layers with efficient addition operations. Collectively, these techniques enable Bi-SSFormer to achieve more effective and efficient HSI classification than prior methods.

To sum up, the contributions of this article are threefold.

1) We propose Bi-SSFormer, a novel ultralightweight spectral–spatial transformer that significantly reduces memory and computation demands for efficient HSI classification.
2) To construct the SS-Block for Bi-SSFormer, we develop the SpeIM and SpaIM to promote the spectral and spatial feature extraction and enhancement with lightweight architecture designs, respectively. Then, InWB is proposed to constrain the network's weights into 1 bit and implement the multiplication-free convolutional and linear layers.
3) Extensive experiments on four benchmark HSI datasets demonstrate the superior classification performance of our method compared with other SOTA models, achieving the lowest memory and computational requirements.

The remainder of this article is organized as follows. Section II reviews the related works. Section III details the proposed Bi-SSFormer and binarization methodology. Section IV presents experimental results and analysis. Finally, Section V provides the conclusion and future research directions.

II. Related Works

A. CNN-Based Models

Benefiting from the inherent advantages of convolution operations, CNNs effectively extract spatial–spectral features from HSIs. Early work [34] utilized 2-D CNNs to jointly exploit spatial and spectral features. To better align with the 3-D cube structure of HSI data, Ben Hamida et al. [35] introduced a novel 3-D CNN model to enable joint spectral–spatial feature extraction for performance enhancement. Furthermore, Roy et al. [41] proposed a hybrid 3-D/2-D CNN model, which was aimed to yield lower complexity and better performance. Nevertheless, CNN-based models suffer from limited representational capacity due to the constrained local receptive fields of the convolutional layers.

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B. Transformer-Based Models

To address CNN limitations, Transformer-based models leverage the long-range modeling capability of self-attention mechanisms to capture global dependencies for improved HSI classification accuracy. For example, Hong et al. [36] proposed SpectralFormer, which learns spectrally local sequence information and employs cross-layer fusion to minimize information loss. Qiu et al. [42] developed a cross-channel dynamic spatial–spectral fusion Transformer for enhanced feature extraction and fusion. Sun et al. [43] presented the spectral–spatial feature tokenization transformer (SSFTT) for feature transformation and high-level learning. Despite these advances, Transformer-based methods exhibit insufficient local feature extraction capability, potentially yielding suboptimal results on specific HSI categories and compromising overall accuracy (OA).

C. CNN–Transformer Hybrid Models

To leverage the complementary strengths of both architectures, CNN–Transformer hybrid models have emerged as a dominant and highly effective paradigm. These models integrate CNNs for robust local feature extraction and Transformers for global context modeling, leading to enhanced classification performance. Early hybrid models demonstrated the power of this combination. For instance, Mei et al. [44] proposed the group-aware hierarchical transformer (GAHT), employing grouped convolution for local spectral–spatial context modeling and hierarchical Transformers for parameter-efficient multiscale learning. Similarly, Roy et al. [45] developed the spectral–spatial morphological attention transformer (MorphFormer), integrating learnable morphological operations with self-attention to enhance feature interactions through adaptive shape-aware fusion.

Driven by the need for real-world deployment on resource-constrained hardware, recent research has increasingly focused on developing lightweight and efficient hybrid models. Zhang et al. [38] presented CAEVT, combining a 3-D convolutional autoencoder for nonlinear dimensionality reduction with lightweight vision transformer blocks for efficient local–global feature extraction, achieving competitive accuracy with significantly reduced FLOPs and parameters. Ma et al. [46] proposed the spatial pooling transformer (SPTNet), utilizing pooling and attention operations to compress spatial dimensions and eliminate redundancy. SPTNet stacked three SPTNets and used only one standard Transformer encoder at last to enhance the spectral–spatial features for HSI classification, balancing the tradeoff between accuracy and model complexity. Zhao et al. [39] introduced the groupwise convolution and Transformer mechanisms to reduce computational and memory requirements while maintaining competitive performance. The proposed groupwise separable MHSA module efficiently extracted the local–global spatial features with reduced computational costs. Zhang et al. [40] proposed the CSCANet, performing cross attention on local–global spatial features while employing group cascade structures for complexity reduction. In addition, they utilized the principal components analysis (PCA) to reduce the input channel number, remarkably reducing the memory and computational requirements for the subsequent model processing.

D. Mamba-Based Models

In recent years, the Mamba-based methods [47], [48] have become the cutting-edge model for efficient and effective HSI classification due to their powerful global modeling capability and linear computational complexity. Wang et al. [49] proposed the local-enhanced Mamba (LE-Mamba), which consisted of a local spatial SSM, a central spectral SSM, and a gated unit, achieving decent performance improvement over previous works. Huang et al. [50] introduced the spectral–spatial Mamba (SS-Mamba), which stacks the Mamba blocks with spectral–spatial feature enhancement modules, allowing the tokens from the respective fields to cooperate. Zhou et al. [51] presented the Mamba-in-Mamba (MiM) architecture, which utilizes the Mamba-cross-scan mechanism, a tokenized encoder, and a weighted fusion module to achieve superior performance in HSI classification tasks. Pan et al. [52] proposed a novel local-to-global Mamba named MambaLG for HSI classification. The spatial modeling module and spectral dynamic perception module were adopted to extract the local and global information, and gate attention was adopted for the feature fusion in an interpretable way.

However, these approaches still represent weights using 32-bit floating-point values and require resource-intensive floating-point multiplications in convolutional and linear layers, indicating substantial potential for further compression. To address this gap, we propose the Bi-SSFormer with weight binarization, advancing efficient yet effective HSI classification.

III. Methodology

A. Overview of Bi-SSFormer

The overall classification framework of the proposed CNN–Transformer hybrid model Bi-SSFormer is illustrated in Fig. 2, comprising three key components: the projection layer, the SS-Blocks, and the classification head.

The spectral channel number of the input varies across different HSI datasets for a given classification model. Following GSC-ViT [39], we first employ a 1×1 convolutional layer to project the variable input channels into a fixed output channel dimension while preserving the original spatial resolution. The SS-Blcoks then extract spectral and spatial features through the SpeIM and the SpaIM, respectively. Regarding SpeIM, we introduce intragroup and cross-group convolutions to enhance spectral band interactions. Instead, for SpaIM, we employ two linear-complexity attention modules, i.e., window attention and grid attention, to capture both local and global spatial dependencies, along with a simple fusion strategy to integrate these features. After processing through multiple SS-Blocks, the global average pooling is applied to reduce spatial dimensions and further refine discriminative features. Finally, the resulting feature map is fed into the classification head for prediction.

The network architecture is designed to be both simple and efficient, leveraging group convolution and efficient

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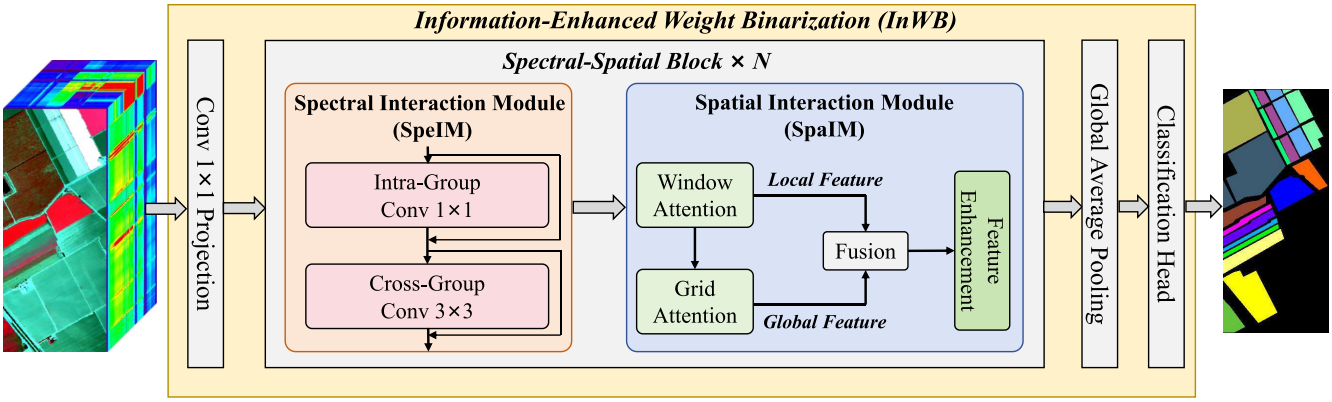


Fig. 2. Overview of the proposed CNN–Transformer hybrid classification model Bi-SSFormer, which consists of the projection layer, the SS-Blocks, the global average pooling, the classification head, and the InWB. The SS-Block contains SpeIM and SpaIM, which, respectively, extract the spectral and spatial features, where N denotes the number of SS-Blocks, set to 2 in this article. The InWB constrains all model weights to 1-bit values (+ 1/−1), achieving significant weight storage and eliminating floating-point multiplications within convolutional and linear layers.

self-attention to minimize computational and memory overhead. Furthermore, we propose an InWB method to quantize floating-point parameters into 1-bit values, significantly reducing memory requirements and eliminating costly floating-point multiplications. The resulting binarized classification model is the proposed Bi-SSFormer.

B. Spectral–Spatial Block

1) SpeIM: Due to the abundant information in the spectral dimension, SpeIM is first applied within each SS-Block of the Bi-SSFormer to promote the information interaction and extract the spectral features. As shown in Fig. 3, the SpeIM has two key components: intragroup convolution and cross-group convolution, where the former is used to extract the spectral features with distinctive group weights, and the latter is used to enable the spectral information interaction with cross-group shared weights.

Intragroup Convolution: Considering the input feature map $\mathbf{X} \in \mathbb{R}^{H\times W\times C_{\text{in}}}$, where H and W are the height and width of $\mathbf{X}$, respectively, and C_{in} is the number of the input channels, the intragroup convolution first divides $\mathbf{X}$ into G groups $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_g, \ldots, \mathbf{X}_G$, where the specific group input is denoted by $\mathbf{X}_g \in \mathbb{R}^{H\times W\times C_{\text{in}}/G}$, with $g \in \{1, \ldots, G\}$. Then, the normal convolutional operation is conducted in each group with shared weights $\mathbf{W}_g \in \mathbb{R}^{C_{\text{out}}/G\times C_{\text{in}}/G\times 1\times 1}$, with $g \in \{1, \ldots, G\}$ and C_{out} being the number of the output channels. This convolution is conducted with a kernel size of 1×1 and the half of the input channel number. This 1×1 convolution allows for linear combinations and dimensionality changes across channels without increasing the spatial receptive field, which aligns with our lightweight design philosophy. After that, the group outputs are concatenated along the channel dimension. In summary, the output $\mathbf{Y} \in \mathbb{R}^{H\times W\times C_{\text{out}}}$ of the intragroup convolution can be computed as follows:

$$\mathbf{Y} = \text{cat}\left[\mathbf{X}_1 * \mathbf{W}_1, \mathbf{X}_2 * \mathbf{W}_2, \ldots, \mathbf{X}_g * \mathbf{W}_g, \ldots, \mathbf{X}_G * \mathbf{W}_G\right] \quad (1)$$

where $*$ denotes the convolutional operation and cat[·] represents the concatenation operation. In addition, to improve the representational power and retain the useful information from the previous stages, we incorporate a residual connection with a channel average pooling (CAP)

$$\mathbf{Y} = \mathbf{Y} + \text{CAP}(\mathbf{X}) \quad (2)$$

where CAP(·) aligns the channel number of input and output feature maps by first reshaping the input as $(H, W, C_{\text{in}}/C_{\text{out}}, C_{\text{out}})$ and conducting the average operation along $C_{\text{in}}/C_{\text{out}}$ dimension. In the end, we conduct the elementwise addition to aggregate these two features. However, the intragroup convolution ignores the interaction between

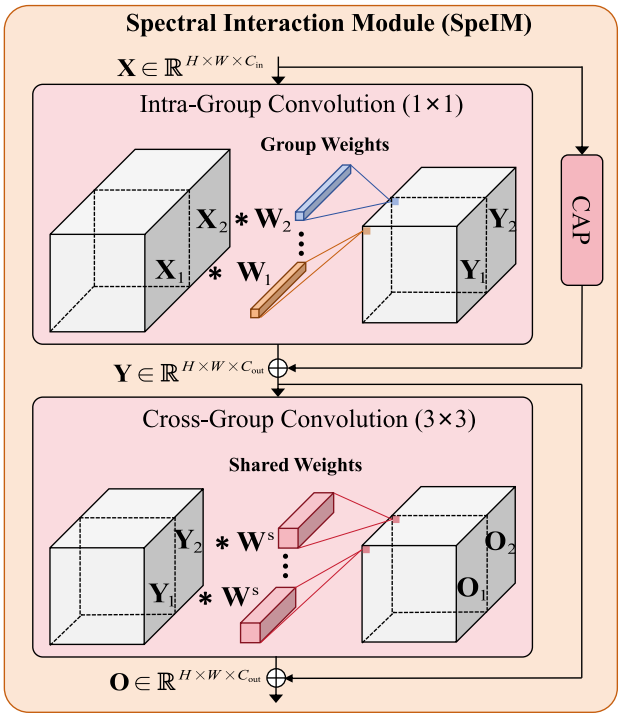


Fig. 3. Illustration of SpeIM, which consists of the intragroup convolution with different group weights and the cross-group convolution with shared weights for effectively extracting the spectral features and promoting the spectral information interaction. CAP is used to align the channel number of input and output feature maps for the residual connections.

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different spectral features in different groups, limiting the representational ability.

Cross-Group Convolution: We propose the cross-group convolution to enable the feature interaction across different groups. Specifically, unlike the individual group weights in the intragroup convolution, the convolution weights with kernel size 3×3 are shared for different group feature maps, represented as $\mathbf{W}^{\mathrm{s}} \in \mathcal{R}^{C_{\mathrm{out}}/G \times C_{\mathrm{out}}/G \times 3 \times 3}$. By using the 3×3 kernel weights, we can simultaneously fuse features from different groups while also incorporating local spatial context. This enables the model to learn a joint spectral–spatial representation, enhancing feature discriminability. Following the intragroup convolutional process, we first split the feature map $\mathbf{Y}$ into the G groups $\mathbf{Y}_1$,..., $\mathbf{Y}_g$,..., $\mathbf{Y}_G$, where the specific group feature is represented by $\mathbf{Y} \in \mathbb{R}^{H \times W \times C_{\mathrm{out}}/G}$ and the group number is consistent with that of intragroup convolution. Then, the output computation process of cross-group convolution can be formulated as follows:

$$\mathbf{O} = \mathrm{cat}\left[\mathbf{Y}_1 * \mathbf{W}^{\mathrm{s}}, \mathbf{Y}_2 * \mathbf{W}^{\mathrm{s}}, \ldots, \mathbf{Y}_g * \mathbf{W}^{\mathrm{s}}, \ldots, \mathbf{Y}_G * \mathbf{W}^{\mathrm{s}}\right] \quad (3)$$

where $\mathbf{O} \in \mathbb{R}^{H \times W \times C_{\mathrm{out}}}$ denotes the output of the SpeIM and $\mathbf{W}^{\mathrm{s}}$ represents the shared weights convolved across the different group feature maps $\mathbf{Y}_g$. Similar to the intra-group convolution, we apply the residual connection to enhance the capacity for effective feature extraction as follows:

$$\mathbf{O} = \mathbf{O} + \mathbf{Y}. \quad (4)$$

Due to the identical channel number of $\mathbf{Y}$ and $\mathbf{O}$, we directly conduct the elementwise addition. The proposed operations jointly extract the spatial–spectral features and effectively improve the representational power of the SpeIM thanks to the cross-group interaction property. Thanks to the weight sharing, the parameter amount of the cross-group convolution is reduced by G times than the intragroup convolution.

Combining the intragroup and cross-group convolution operations, the parameter number and FLOPs of SpeIM can be calculated by $(C_{\mathrm{in}}C_{\mathrm{out}})/G + (9C_{\mathrm{out}}^2)/G^2$ and $(\mathrm{HWC}_{\mathrm{in}}C_{\mathrm{out}})/G + (9HWC_{\mathrm{out}}^2)/G$, respectively. Compared with the standard convolution with parameter amount $9C_{\mathrm{in}}C_{\mathrm{out}}$ and $9HWC_{\mathrm{in}}C_{\mathrm{out}}$ FLOPs, we can enhance the feature extraction capability of the convolutional layers effectively by saving $1/(9G) + C_{\mathrm{out}}/(G^2C_{\mathrm{in}})$ memory requirement and $1/(9G) + C_{\mathrm{out}}/(\mathrm{GC}_{\mathrm{in}})$ computational cost.

2) SpaIM: To derive the discriminative features for highly accurate prediction results, the ability to effectively extract spatial features is of great relevance in HSI classification. Hong et al. [36] and Roy et al. [45] proposed various kinds of attention mechanisms to model the long-range spatial dependencies between different locations within the feature maps. However, these methods either use quadratic-complexity self-attention or design complicated architectures for the local–global feature extraction and integration. To solve these problems, we propose an innovative SpaIM to conduct the local–global attention process efficiently and effectively. As demonstrated in Fig. 4, the SpaIM contains four components, which are the window attention, grid attention, fusion, and feature enhancement modules.

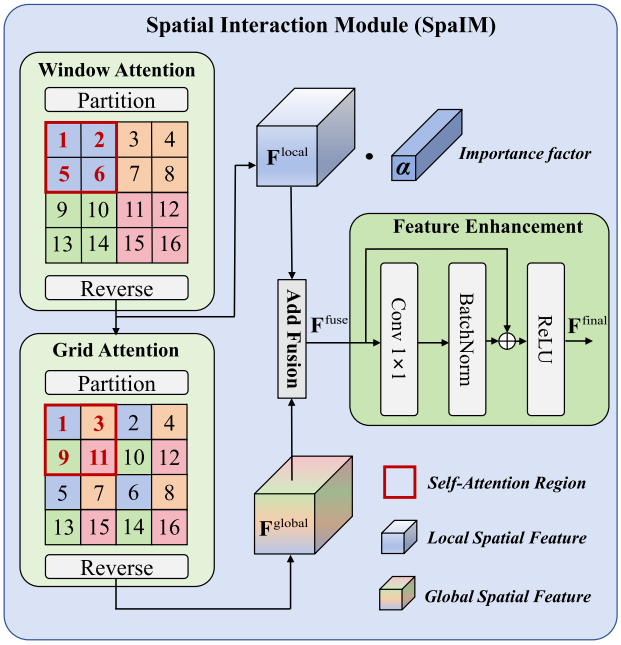


Fig. 4. Illustration of SpaIM when the feature map size is set to 4×4 with partition size $P = 2 \times 2$, which applies the window and grid attention operations for local and global feature extraction, respectively. Different colors with various index numbers denote the self-attention targets within the computational regions. Then, we use the simple yet effective addition to fuse local–global features and propose the feature enhancement module for further information refinement.

Window Attention: The window attention module aims to extract the local information of the input feature map. Specifically, we first represent the input as $\mathbf{O} \in \mathbb{R}^{H \times W \times C_{\mathrm{out}}}$, which is also the output of the SpeIM. Then, to reduce the computational complexity, we partition the input feature map into multiple equal-sized windows with a reshape operation as follows:

$$(H, W, C_{\mathrm{out}}) \rightarrow \left(\frac{H}{P} \times \frac{W}{P}, P \times P, C_{\mathrm{out}}\right) \quad (5)$$

where the hyperparameter P controls the window size and we conduct the vanilla MHSA [53] within the partitioned window feature map $P \times P$ to implement the local feature interaction. As demonstrated in Fig. 4, the self-attention operation is conducted within the red region, consisting of the local features including the 1st, 2nd, 5th, and 6th visual tokens. Afterward, we reverse the output local feature $\mathbf{F}^{\mathrm{local}}$ into the original shape (H, W, C_{out}) for the subsequent process.

Grid Attention: The following grid attention module aims to extract the global information with dilated feature mixing [54]. In detail, considering the input with shape (H, W, C_{out}), we first conduct the grid partition reshape operation as follows:

$$(H, W, C_{\mathrm{out}}) \rightarrow \left(P \times P, \frac{H}{P} \times \frac{W}{P}, C_{\mathrm{out}}\right) \quad (6)$$

where we use the same hyperparameter P to control the grid window size $H/P \times W/P$, which is adaptive according to the spatial size of the input feature map. Then, we also use the vanilla MHSA module to conduct the self-attention within the grid window feature map, enabling the global feature interaction. As shown in Fig. 4, the self-attention operation is

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conducted within the red region which consists of the mixing global features, including the 1st, the 3rd, the 9th, and the 11th visual tokens from different partitioned windows. Afterward, as in the window attention, we reshape the output global feature $\mathbf{F}^{\text{global}}$ matching the size (H, W, C_{out}).

In addition, thanks to the partition-then-self-attention strategy, it is worth noting that both window and grid attention only have a linear complexity $\text{HW} \cdot P^2 \cdot C_{\text{out}}$ compared with the vanilla global attention conducted on the entire feature map with quadratic complexity $(\text{HW})^2 \cdot C_{\text{out}}$, strongly saving computational budgets by $P^2/(\text{HW})$.

Fusion Module: As reported in the previous works [39], [40], the local and global features are both important for classification performance. To this end, we propose a simple yet effective fusion module to integrate these two kinds of features, enhancing the effectiveness of the SpaIM. After obtaining the local feature $\mathbf{F}^{\text{local}} \in \mathbb{R}^{H\times W\times C_{\text{out}}}$ and global feature $\mathbf{F}^{\text{global}} \in \mathbb{R}^{H\times W\times C_{\text{out}}}$ with the window and grid attention modules, respectively, we introduce a channelwise importance factor $\boldsymbol{\alpha} \in \mathbb{R}^{C_{\text{out}}}$ to balance the importance between the local and global information and use the elementwise addition to fuse the weighted features as follows:

$$\mathbf{F}^{\text{fuse}} = \mathbf{F}^{\text{local}} \odot \boldsymbol{\alpha} + \mathbf{F}^{\text{global}} \tag{7}$$

where $\odot$ represents the elementwise multiplication with channelwise broadcasting, which emphasizes that $\boldsymbol{\alpha}$ is expanded along the height and width dimensions to match the shape of $\mathbf{F}^{\text{local}}$ during the scaling process. The resulting fused feature output $\mathbf{F}^{\text{fuse}} \in \mathbb{R}^{H\times W\times C_{\text{out}}}$ encapsulates both local and global interaction features, effectively combining detailed local information with comprehensive global context. This fusion mechanism not only allows the model to capture both fine-scale and large-scale patterns but also improves the overall representational power of SpaIM, leading to enhanced performance in classification tasks.

Feature Enhancement Module: To further enhance the diversity of local–global features and extract the high-level semantic information, we introduce the feature enhancement module to process the fused output features as follows:

$$\mathbf{F}^{\text{final}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times 1}\left(\mathbf{F}^{\text{fuse}}\right)\right) + \mathbf{F}^{\text{fuse}}\right) \tag{8}$$

where $\text{ReLU}(\cdot)$ is the rectified linear unit activation function, $\text{BN}(\cdot)$ denotes the batch-norm layer, and $\text{Conv}_{1\times 1}(\cdot)$ represents the convolutional layer with kernel size 1×1. In addition, we introduce the residual connection to help improve the representational power and ease the training.

The above-described components constitute the SS-Block of the proposed Bi-SSFormer. By stacking this block N times and gradually downsampling the channel number, we derive highly expressive features for HSI classification. In this article, we set N to 2, which means that the spectral channel number is reduced by a factor of 4.

C. Information-Enhanced Weight Binarization

In Sections III-A and III-B, we presented a lightweight CNN–Transformer hybrid network, which consists of the efficient groupwise spectral interactive convolution process and linear-complexity local–global information interaction modules. Despite its low model complexity, there is still room for further model compression by constraining the bit width of weight representations. To this end, we propose the InWB method to binarize the 32-bit weights of the lightweight network into only 1 bit, significantly saving memory and replacing the floating-point multiplications with simple additions.

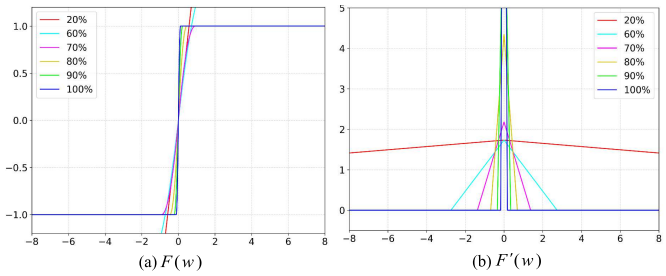


Fig. 5. Proposed gradient estimator in the InWB. (a) Approximate function and (b) its derivative function through the training process.

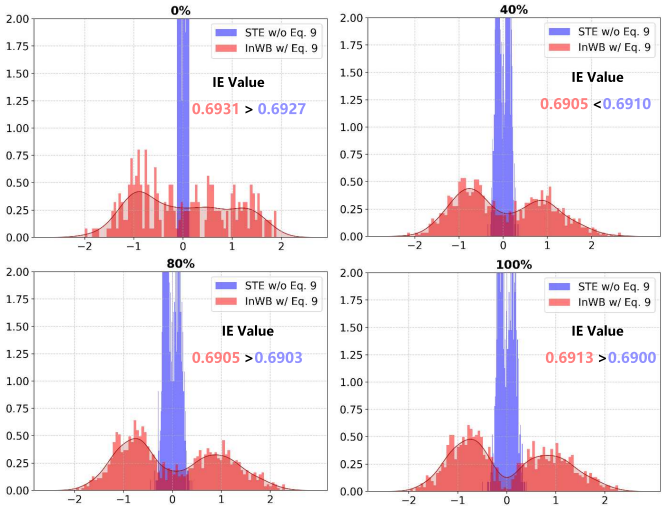


Fig. 6. Visualization of the weight distributions of the last linear layer in Bi-SSFormer on the Salinas dataset at different training stages, including 0%, 40%, 80%, and 100%. We compare the proposed InWB w/ standardization [using (9)] with the STE w/o standardization and report the changed IE Value through the training process. The IE Value is calculated by $-p\ln p - (1-p)\ln(1-p)$, where p refers to the probability of weights binarized into + 1.

Specifically, for the sake of simplicity, we use $\mathbf{W} \in \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}}$ to represent the parameters of an arbitrary convolutional or a linear layer within the proposed model. Based on the conclusion derived by IRNet [55], the binarized weights take the larger information entropy value (IE Value) when they are evenly distributed. Therefore, to enhance the information on binarization, the forward process of the InWB is formulated as follows:

$$\widehat{\mathbf{W}} = \frac{\mathbf{W} - \mu}{\sigma} \tag{9}$$

$$\mathbf{B}_{\text{W}} = \text{sign}\left(\widehat{\mathbf{W}}\right) \tag{10}$$

where $\widehat{\mathbf{W}}$ is the standardized weight matrix, obtained by subtracting the mean value μ of $\mathbf{W}$ and dividing by the standard deviation σ of $\mathbf{W}$, $\mathbf{B}_{\text{W}}$ is the binarized weight, and $\text{sign}(\cdot)$ is

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the sign function outputting −1 for negative inputs and +1 for positive inputs. In this way, the information about the weight is enhanced, and the parameter can be saved with only 1 bit. In addition, the floating-point multiplications are eliminated and replaced by the simple sign adjustment operation.

For the backward process, to solve the nondifferentiable problem of the function sign(·), previous works usually use the straight-through estimator [56] or the fixed quadratic function [57] to approximate the gradient for model training. However, this method may lead to a large gradient mismatch through the training process, resulting in information loss and quantization errors.

To alleviate this problem, we propose the adaptive gradient estimator derived from the normal quadratic function, imposing strong gradients on the weights to guide their sign determination, while gradually approximating the function sign(·) to alleviate the gradient mismatch issue. Mathematically, we have

$$F(w) = \begin{cases} r\left(-\text{sign}(w)\dfrac{3t^2w^2}{4} + \sqrt{3}tw\right), & \text{if } |w| < \dfrac{2\sqrt{3}}{3t} \\ r\,\text{sign}(w), & \text{otherwise} \end{cases} \tag{11}$$

where $|\cdot|$ is the absolute value and r and t are used to adjust the shape of the InWB estimator during the training and computed as follows:

$$t = 10^{\frac{3e}{E}-2}, \quad r = \max\left(\frac{1}{t}, 1\right) \tag{12}$$

where e and E denote the current training epoch and the total epoch number, respectively, and $\max(\cdot,\cdot)$ is the maximum operator. The derivative of the InWB estimator is as follows:

$$F'(w) = \begin{cases} r\left(-\text{sign}(w)\dfrac{3t^2w}{2} + \sqrt{3}t\right), & \text{if } |w| < \dfrac{2\sqrt{3}}{3t} \\ 0, & \text{otherwise.} \end{cases} \tag{13}$$

To more intuitively explain the proposed method, we visualize $F(w)$ and $F'(w)$ during the training process in Fig. 5. At the beginning of the training phase, the gentler slope of the gradient curve in the early training stages (small t and large r) provides a more robust gradient signal, encouraging weights to effectively explore between −1 and +1 and avoiding premature convergence in poor local minima due to vanishing gradients. As training progresses, the curve steepens, better approximating the true gradient of the sign function, which helps the weights to stabilize and converge to their final binary states, thus alleviating the gradient mismatch problem throughout the training process.

Moreover, as demonstrated in Fig. 6, the weight distribution evolves from a randomly initialized shape into a distinct bimodal distribution concentrated around +1 and −1 under the guidance of our InWB method. It can be found that the IE Value of our method is better retained than the STE method without standardization (9) during the optimization process. After training, the weights learned by our method obtain a larger information entropy. Therefore, we can say that thanks to the balance and standardization operations applied before binarization along with the proposed InWB, the information entropy of the binarized weights in the Bi-SSFormer is optimized.

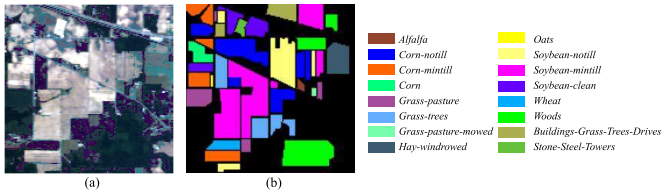


Fig. 7. (a) False color image (using 29th, 19th, and 9th bands) and its (b) ground-truth counterpart for the Indian Pines dataset. The rectangular blocks with different colors represent different categories.

TABLE I
COMPOSITION OF SAMPLES IN THE TRAINING SET, VALIDATION SET, AND TESTING SET FOR THE INDIAN PINES DATASET

Class No.	Class Name	Train	Val	Test
1	Alfalfa	3	2	41
2	Corn-notill	72	71	1285
3	Corn-mintill	42	41	747
4	Corn	12	12	213
5	Grass-pasture	24	24	435
6	Grass-trees	36	37	657
7	Grass-pasture-mowed	1	2	25
8	Hay-windrowed	24	24	430
9	Oats	1	1	18
10	Soybean-notill	49	48	875
11	Soybean-mintill	122	123	2210
12	Soybean-clean	30	29	534
13	Wheat	10	10	185
14	Woods	63	63	1139
15	Buildings-Grass-Trees-Drives	19	20	347
16	Stone-Steel-Towers	4	5	84
	Total	512	512	9225

IV. EXPERIMENTS

A. Experimental Settings

1) HSI Datasets: In this article, we evaluate the proposed method Bi-SSFormer on four benchmark hyperspectral datasets, which are described as follows.

1) *Indian Pines:* Acquired by the NASA AVIRIS instrument over Indian Pines in 1992, this dataset features a 145 × 145 spatial size with 220 spectral bands. After removing 20 water-absorptive bands, 200 bands remain. It contains 16 land cover classes, with 10 249 labeled samples. The dataset is divided into 5% for training, 5% for validation, and 90% for testing as detailed in Table I, and its false color composite and ground-truth map are shown in Fig. 7.
2) *Salinas:* Captured by AVIRIS over Salinas Valley, California, this dataset has a 512 × 217 spatial resolution and 224 spectral bands. After excluding 20 water bands, 204 bands are used. It includes 16 crop classes with 54 329 samples, where the dataset is divided into 1% for training, 1% for validation, and 98% for testing as detailed in Table II, and its false color composite and ground-truth map are shown in Fig. 8.
3) *WHU-Hi LongKou:* Collected via uncrewed aerial vehicle (UAV) in Hubei, China, this dataset has a

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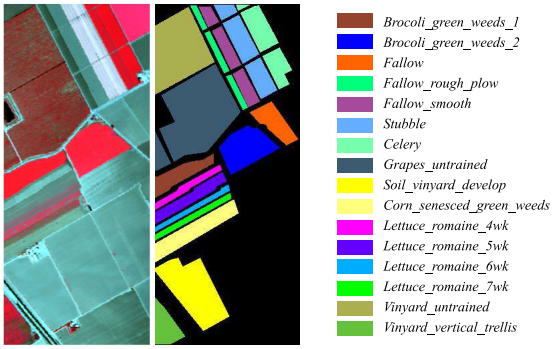


Fig. 8. (a) False color image (using 50th, 30th, and 20th bands) and its (b) ground-truth counterpart for the Salinas dataset. The rectangular blocks with different colors represent different categories.

TABLE II

COMPOSITION OF SAMPLES IN THE TRAINING SET, VALIDATION SET, AND TESTING SET FOR THE SALINAS DATASET

Class No.	Class Name	Train	Val	Test
1	Brocoli-green-weeds-1	20	20	1969
2	Brocoli-green-weeds-2	37	37	3652
3	Fallow	20	20	1936
4	Fallow-rough-plow	14	14	1366
5	Fallow-smooth	27	27	2624
6	Stubble	40	39	3880
7	Celery	36	36	3507
8	Grapes-untrained	113	112	11046
9	Soil-vinyard-develop	62	62	6079
10	Corn-senesced-green-weeds	33	33	3212
11	Lettuce-romaine-4wk	11	10	1047
12	Lettuce-romaine-5wk	20	19	1888
13	Lettuce-romaine-6wk	9	9	898
14	Lettuce-romaine-7wk	10	11	1049
15	Vinyard-untrained	73	72	7123
16	Vinyard-vertical-trellis	18	18	1771
	Total	543	539	53047

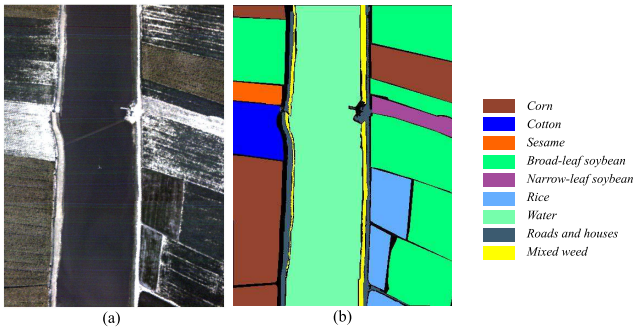


Fig. 9. (a) False color image (using 29th, 19th, and 9th bands) and (b) its ground-truth counterpart for the WHU-Hi LongKou dataset. The rectangular blocks with different colors represent different categories.

550 × 400 spatial resolution and 270 bands spanning from 0.4 to 1.0 μm. It includes nine classes with 204 542 samples, where the dataset is divided into 0.2% for training, 0.2% for validation, and 99.6% for testing as detailed in Table III, and its false color composite and ground-truth map are shown in Fig. 9.

4) *Botswana:* Acquired by the NASA EO-1 satellite over the Okavango Delta, Botswana, in 2001, this dataset has

TABLE III

COMPOSITION OF SAMPLES IN THE TRAINING SET, VALIDATION SET, AND TESTING SET FOR THE WHU-HI LONGKOU DATASET

Class No.	Class Name	Train	Val	Test
1	Corn	69	69	34373
2	Cotton	17	16	8341
3	Sesame	6	6	3019
4	Broad-leaf soybean	127	126	62959
5	Narrow-leaf soybean	9	8	4134
6	Rice	24	23	11807
7	Water	134	134	66788
8	Roads and houses	15	14	7095
9	Mixed weed	11	10	5208
	Total	412	406	203724

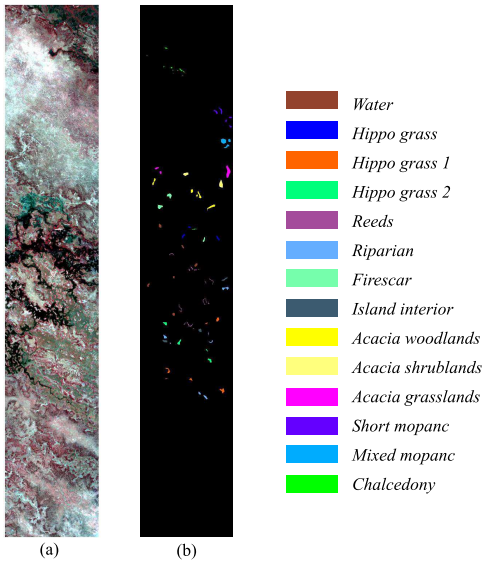


Fig. 10. (a) False color image (using 29th, 19th, and 9th bands) and (b) its ground-truth counterpart for the Botswana dataset. The rectangular blocks with different colors represent different categories.

a 1476 × 256 spatial size and 242 bands. After removing noise bands, 145 bands remain. It comprises 14 land cover classes with 3248 samples, where the dataset is divided into 5% for training, 5% for validation, and 90% for testing as detailed in Table IV, and its false color composite and ground-truth map are shown in Fig. 10.

2) Evaluation Metrics: The model performance and complexity are assessed using the following metrics.

1) *OA:* Percentage of correctly classified pixels.
2) *Average Accuracy (AA):* Mean accuracy across all classes.
3) *Kappa Coefficient ($\kappa \times 100$):* Measuring classification consistency beyond random chance.
4) *Params:* The number of bytes required to store model parameters.
5) *Floating-Point Operation (FLOPs):* The FLOPs in a single forward of the Bi-SSFormer when using a single-HSI input patch.

3) Implementation Details: The proposed Bi-SSFormer is implemented using PyTorch on a single PC with an NVIDIA

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TABLE IV
COMPOSITION OF SAMPLES IN THE TRAINING SET, VALIDATION SET, AND TESTING SET FOR THE BOTSWANA DATASET

Class No.	Class Name	Train	Val	Test
1	Water	14	13	243
2	Hippo-grass	5	5	91
3	Hippo grass-1	13	12	226
4	Hippo grass-2	11	10	194
5	Reeds	14	13	242
6	Riparian	14	13	242
7	Firescar	13	13	233
8	Island interior	10	10	183
9	Acacia shrublands	16	15	283
10	Acacia shrublands	13	12	223
11	Acacia grasslands	15	15	275
12	Short mopanc	9	9	163
13	Mixed mopanc	14	13	241
14	Chalcedony	5	5	85
	Total	166	158	2924

RTX 3090 GPU. We apply AdamW as the default optimizer and set the initial learning rate to 0.001 with a weight decay of 0.05. The batch size is set to 128, and the total training epoch is 200. For every pixel in the ground-truth map that has a valid class label, we extract a surrounding patch of size 8×8 as the input by default. The label for this entire patch is unequivocally assigned as the label of that central reference pixel. Then, we use the projection layer to align the input channel number to 256 for further processing. In addition, we use data augmentation techniques, including horizontal/vertical flipping and random rotation, to mitigate overfitting. We run each experiment ten times reporting the mean and standard deviation values.

B. Ablation Study

To explore the effectiveness of the proposed method, we carried out a comprehensive ablation study on the four HSI datasets, where SpeIM, SpaIM, and InWB refer to the proposed spectral interaction module, spatial interaction module, and information-enhanced weight binarization methods. The baseline model is built based on the structure of the Bi-SSFormer, which replaces SpeIM and SpaIM with the normal convolutional layer and vanilla Transformer block, respectively. This baseline is inherently more complex, with a larger number of parameters and a higher computational cost. Table V presents the classification results, memory, and computational complexity of our method when breakdown experiments are conducted.

When applying SpeIM to the baseline model, the model performance obtains the 0.12%, 0.01%, 0.03%, and 0.04% OA gains on four datasets, demonstrating the effectiveness of the intragroup and cross-group convolution to promote the spectral feature interaction. At the same time, the memory requirements are reduced by 130.30, 130.30, 149.18, and 130.30 kB while the FLOPs are decreased by 0.77M, 0.77M, 1.67M, and 0.77M, showing its lightweight advantages over the direct convolutional operation. When solely using the SpaIM, due to the linear-complexity attention paradigm and simple fusion strategy, the memory and computation budgets of the model on the four datasets are saved by almost 50%. Thanks to the effective local–global attention structure, the proposed SpaIM improves the OA of the baseline model by 0.71%, 0.27%, 0.21%, and 0.28% on these datasets. Then, combining SpeIM and SpaIM, the derived model achieves the best model performance and the least model complexity. Moreover, by applying the InWB on the baseline model, the memory and computational requirements decrease by over $30\times$ and $4\times$, respectively. However, due to the constrained representation of weights, the classification performance is also degenerated by 0.86%, 0.17%, and 0.07% on Indian Pines, WHU-Hi-LongKou, and Botswana datasets, respectively, compared it with the baseline with full-precision weights. To alleviate this problem, we combine the InWB with the SpeIM and SpaIM to derive the final Bi-SSFormer, achieving the best tradeoff between the model performance and complexity across different datasets.

1) Study on SpeIM: To study the effect of the proposed SpeIM, we compare it with the groupwise separable convolution (GSC) [39] under different group numbers and datasets in Table VI. The GSC utilizes two consecutive groupwise convolutional layers to extract the spectral features. Compared with GSC, as demonstrated in Table VI, the proposed strategy achieves a better OA across the four datasets with less memory costs, which benefits from the shared weights within the cross-group convolution and a stronger representation of the spectral feature interaction. Moreover, the best choice of group number for SpeIM is dataset dependent. For example, the highest OA for Indian Pines and Salinas is obtained when the group number is 2, while the best performance for WHU-Hi LongKou and Botswana is achieved when the group number is set to 16 and 8, respectively.

From this table, we can observe that the extent of memory savings gradually decreases as the group number increases, especially during the transition from 8 to 16. Therefore, to build a more lightweight HSI model, we select the best group number for SpeIM from 8 and 16 based on the OA metric for the corresponding datasets.

2) Study on SpaIM: To explore the influence of the proposed SpaIM, we compare it with the vanilla MHSA and GMHSA in Table VII, where the former is the MHSA with quadratic complexity and the latter is the group MHSA proposed in GSC-ViT [39]. As shown in the table, the SpaIM with fusion achieves the best OA with $96.55 \pm 0.43\%$, $96.72 \pm 0.64\%$, and $98.86 \pm 0.23\%$ for Indian Pines, Salinas, and WHU-Hi LongKou datasets, improving the classification performance significantly from 0.02% to 0.62%. These results showcase that our method can better model local and global features, facilitating the spatial information interaction for representational power enhancement. Moreover, we also explore the effect of the fusion module on the classification performance by eliminating it from the SpaIM. It can be noted that a dataset-based degradation of the OA from 0.19% to 3.36% for three out of four test cases. Instead, just for the Botswana dataset, the OA is improved by 0.11% when the fusion module is not applied. The reason for this behavior

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TABLE V

ABLATION STUDIES ON THE FOUR HSI DATASETS USING THE PROPOSED METHOD, INCLUDING SPEIM, SPAIM, AND INWB. BEST RESULTS ARE IN BOLDFACE.KB AND M STAND FOR KILOBYTE AND MILLION, RESPECTIVELY

Exp Number	SpeIM	SpaIM	InWB	Indian Pines			Salinas			WHU-Hi-LongKou			Botswana		
				Params (KB)	FLOPs (M)	OA (%)	Params (KB)	FLOPs (M)	OA (%)	Params (KB)	FLOPs (M)	OA (%)	Params (KB)	FLOPs (M)	OA (%)
1				1137.92	18.14	96.73±0.42	1092.48	17.42	96.64±0.71	1257.34	20.07	98.56±0.29	1031.55	16.45	98.54±0.58
2	✓			1007.62	17.37	96.85±0.55	962.18	16.65	96.65±0.41	1108.16	18.40	98.59±0.36	901.25	15.68	98.58±0.47
3		✓		662.02	10.58	**97.44±0.39**	567.04	9.08	96.91±0.38	830.98	13.30	98.77±0.23	506.11	8.11	98.82±0.50
4	✓	✓		531.72	9.82	97.36±0.56	436.74	8.31	**97.01±0.53**	681.79	11.63	98.85±0.30	375.81	7.34	**98.94±0.52**
5			✓	35.56	9.07	95.87±0.80	34.14	8.71	96.80±0.63	39.29	10.04	98.39±0.33	32.24	8.23	98.47±0.53
6	✓	✓	✓	**16.62**	**4.91**	96.55±0.43	**13.65**	**4.15**	96.72±0.64	**21.31**	**5.82**	**98.86±0.23**	**11.74**	**3.67**	98.76±0.30

TABLE VI

ABLATION STUDY ON THE EFFECTIVENESS OF THE PROPOSED SPEIM. BEST RESULTS ARE IN BOLDFACE. KB STANDS FOR KILOBYTE

Group Number	Method	Indian Pines			Salinas			WHU-Hi-LongKou			Botswana		
		Params (KB)	AA (%)	OA (%)	Params (KB)	AA (%)	OA (%)	Params (KB)	AA (%)	OA (%)	Params (KB)	AA (%)	OA (%)
2	GSC	29.70	93.01±1.58	96.52±0.74	26.73	98.12±0.51	96.59±0.67	34.98	95.22±1.44	98.55±0.52	24.82	98.54±0.71	98.69±0.58
	SpeIM	23.94	93.77±1.55	**96.76±0.39**	20.97	**98.23±0.40**	**96.91±0.60**	29.22	95.68±1.27	98.74±0.33	19.06	98.69±0.61	98.75±0.57
4	GSC	22.66	93.61±1.59	96.45±0.46	19.69	97.83±0.45	96.54±0.54	27.94	95.79±1.34	98.64±0.37	17.78	98.52±0.84	98.65±0.60
	SpeIM	18.34	93.76±1.40	96.74±0.60	15.37	98.08±0.47	96.75±0.49	23.62	95.11±1.98	98.60±0.41	13.46	98.60±0.55	98.76±0.43
8	GSC	19.14	93.60±1.47	96.37±0.71	16.17	97.79±0.61	96.57±0.72	24.42	96.01±0.81	98.78±0.39	14.26	98.67±0.46	98.71±0.42
	SpeIM	16.62	**94.06±1.73**	96.55±0.43	13.65	98.04±0.50	96.72±0.64	21.90	95.92±0.56	98.65±0.38	11.74	**98.74±0.38**	**98.76±0.30**
16	GSC	17.38	93.11±2.10	96.37±0.65	14.41	97.76±0.54	96.54±0.46	22.66	95.78±1.70	98.66±0.47	12.50	98.70±0.55	98.74±0.41
	SpeIM	**16.03**	92.35±1.62	96.32±0.68	**13.06**	97.99±0.47	96.72±0.62	**21.31**	**96.09±0.68**	**98.86±0.23**	**11.15**	98.70±0.47	98.76±0.46

TABLE VII

ABLATION STUDY ON THE EFFECTIVENESS OF THE PROPOSED SPAIM. BEST RESULTS ARE IN BOLDFACE

Method	Indian Pines			Salinas			WHU-Hi-LongKou			Botswana		
	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
Vanilla MHSA	95.99±0.58	92.53±1.89	95.43±0.65	96.70±0.66	**98.06±0.50**	96.32±0.74	98.40±0.60	94.89±1.75	97.90±0.78	98.14±0.60	98.12±0.59	97.99±0.65
GMHSA	95.72±0.43	90.80±2.56	95.12±0.48	96.62±0.41	97.97±0.46	96.25±0.46	98.46±0.33	94.80±1.49	97.97±0.43	98.24±0.68	98.05±0.79	98.09±0.74
SpaIM w/o Fusion	93.19±4.71	88.02±7.81	92.23±5.38	96.32±0.76	92.63±1.71	95.81±0.86	98.67±0.30	95.61±1.43	98.26±0.39	**98.87±0.27**	**98.86±0.30**	**98.78±0.30**
SpaIM w/ Fusion	**96.55±0.43**	**94.06±1.73**	**96.06±0.49**	**96.72±0.64**	98.04±0.50	**96.35±0.71**	**98.86±0.23**	**96.09±0.68**	**98.50±0.30**	98.76±0.30	98.74±0.38	98.65±0.32

TABLE VIII

COMPARISONS OF THE CLASSIFICATION PERFORMANCE AND COMPLEXITY WITH THE DGF AND CAF STRATEGIES ON INDIAN PINES AND SALINAS DATASETS

Method	Indian Pines			Salinas		
	OA(%)	Params (KB)	FLOPs (M)	OA(%)	Params (KB)	FLOPs (M)
SpaIM w/ DGF	**96.62±0.55**	21.74	6.22	**96.77±0.74**	18.77	5.46
SpaIM w/ CAF	95.54±0.68	27.05	7.58	96.17±0.59	24.08	6.82
SpaIM w/ Our Fusion	96.55±0.43	**16.62**	**4.91**	96.72±0.64	**13.65**	**4.15**

could be related to the fact that this dataset requires more global modeling capabilities to extract long-distance features, while local features probably harm the final feature quality, as shown in Fig. 10.

Moreover, to adequately justify our choice over more complex feature fusion methods, as illustrated in Fig. 11, we implement and evaluate two alternatives.

1) *Dynamic Gated Fusion (DGF):* This method concatenates the local and global features and uses a 1×1 convolution with a sigmoid function to generate a spatially aware gate map $\mathbf{G}$, which then adaptively combines the two feature maps.
2) *Cross-Attention Fusion (CAF):* This module utilizes a cross-attention mechanism to enhance a more complex feature interaction. The local feature map generates the query ($\mathbf{Q}$) tensor, which interacts with the key ($\mathbf{K}$) and value ($\mathbf{V}$) tensors derived from the global feature map. This process allows each local position to dynamically aggregate the most relevant global context.

where we compare them with our method on Indian Pines and Salinas datasets and report the OA, memory, and computational requirements in Table VIII. As shown in this table, the more complex CAF with cross-attention interaction achieves the lowest OA with the most parameters and computation cost. Although the DGF achieves the best accuracy with the dynamic gating fusion, its performance improvement over our method is marginal, with gains of 0.07% and 0.05% OA on Indian Pines and Salinas datasets, respectively. In contrast, our proposed fusion method achieves the optimal balance between performance and complexity, where its OA is competitive with DGF while saving up to $1.38\times$ and $1.32\times$ memory and computational requirements, respectively. These results validate that, compared with the more sophisticated feature interaction mechanisms, the proposed simple fusion method is more suitable for building an ultralightweight binary network for effective and efficient HSI classification.

3) Study on InWB: To show the superiority of the proposed InWB on the weight binarization, we compare our method with the vanilla BNN [58] and other SOTA binary neural networks, including OvSW [59] and BiPer [60] in Table IX. It can be observed that our method achieves a remarkable classification performance improvement over the vanilla binary method BNN, obtaining an 8.83% OA gain on Indian Pines, a 1.85% gain on Salinas, a 1.85% gain on WHU-Hi LongKou, and a 1.94% gain on Botswana. OvSW and BiPer also exhibit the effectiveness of reducing the quantization error, enhancing the OA of the baseline method from 0.76% to 7.24% across the four HSI datasets. Despite this, thanks to the information-enhanced forward and backward process of our weight binarization strategy, the InWB can reduce the information loss in convolution and linear operations and alleviate the gradient mismatch for more effective training with binarized weights, leading to the best results on the OA, AA, and kappa coefficient metrics.

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TABLE IX

ABLATION STUDY ON THE EFFECTIVENESS OF THE PROPOSED INWB. BEST RESULTS ARE IN BOLDFACE

Method	Indian Pines			Salinas			Whu-Hi LongKou			Botswana		
	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
BNN	87.72±1.72	79.15±3.72	85.97±1.97	94.87±0.82	96.07±1.68	94.29±0.91	97.01±0.44	89.98±1.75	96.06±0.58	96.82±0.88	96.77±1.36	96.55±0.96
OvSW	93.06±0.83	86.30±3.28	92.08±0.94	95.58±0.44	96.97±0.63	95.08±0.49	97.77±0.31	91.73±1.27	97.07±0.41	98.34±0.47	98.29±0.54	98.21±0.50
BiPer	94.96±0.71	91.26±2.40	94.26±0.81	95.80±0.55	97.14±0.64	95.33±0.61	98.01±0.47	94.13±1.82	97.39±0.62	97.88±0.67	97.57±1.02	97.71±0.73
InWB	**96.55±0.43**	**94.06±1.73**	**96.06±0.49**	**96.72±0.64**	**98.04±0.50**	**96.35±0.71**	**98.86±0.23**	**96.09±0.68**	**98.50±0.30**	**98.76±0.30**	**98.74±0.38**	**98.65±0.32**

TABLE X

STUDY ON THE EFFECT OF DIFFERENT PATCH SIZES AND PARTITION SIZES ON THE CLASSIFICATION PERFORMANCE AND COMPUTATIONAL COMPLEXITY OF THE PROPOSED MODEL. BEST RESULTS ARE IN BOLDFACE. M STANDS FOR MILLION

Patch size (Partition size)	Indian Pines			Salinas			WHU-Hi LongKou			Botswana		
	OA(%)	AA(%)	FLOPs(M)	OA(%)	AA(%)	FLOPs(M)	OA(%)	AA(%)	FLOPs(M)	OA(%)	AA(%)	FLOPs(M)
14×14 (7×7)	96.25±0.50	91.86±2.49	15.03	97.87±0.18	98.00±0.33	12.72	98.40±0.50	94.60±1.86	17.81	98.29±0.80	98.14±0.92	11.24
14×14 (2×2)	96.10±0.48	91.97±2.24		**98.49±0.46**	**98.92±0.48**		98.70±0.23	96.04±0.74		98.75±0.66	98.74±0.77	
12×12 (6×6)	96.21±0.79	92.37±2.14	11.04	97.70±0.31	98.52±0.19	9.34	98.42±0.38	95.13±1.24	13.09	98.77±0.43	98.80±0.37	8.26
12×12 (4×4)	96.13±0.79	92.97±1.81		98.02±0.27	98.71±0.29		98.56±0.33	**96.30±0.94**		98.65±0.71	98.73±0.66	
12×12 (3×3)	96.30±0.68	91.81±1.43		97.88±0.31	98.47±0.37		98.72±0.19	95.83±0.58		98.84±0.37	**98.93±0.31**	
12×12 (2×2)	96.59±0.69	92.10±2.13		97.99±0.58	98.70±0.41		98.69±0.26	95.54±1.07		98.74±0.67	98.70±0.81	
10×10 (5×5)	96.47±0.57	93.11±1.20	7.67	97.37±0.53	98.44±0.35	6.49	98.60±0.27	95.46±1.04	9.09	98.39±0.70	98.33±0.76	5.73
10×10 (2×2)	**96.80±0.51**	93.02±2.10		97.43±0.38	98.31±0.32		98.80±0.22	96.06±0.62		**98.94±0.35**	98.86±0.35	
8×8 (4×4)	96.01±0.40	91.82±1.74	**4.91**	96.45±0.64	97.82±0.59	**4.15**	98.50±0.29	95.61±0.63	**5.82**	98.61±0.66	98.45±0.85	**3.67**
8×8 (2×2)	96.55±0.43	**94.06±1.73**		96.72±0.64	98.04±0.50		**98.86±0.23**	96.09±0.68		98.76±0.30	98.74±0.38	

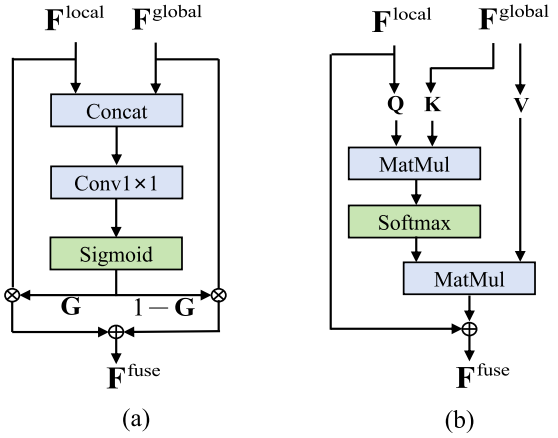


Fig. 11. Illustration of the more complex feature fusion strategies, including (a) DGF and (b) CAF.

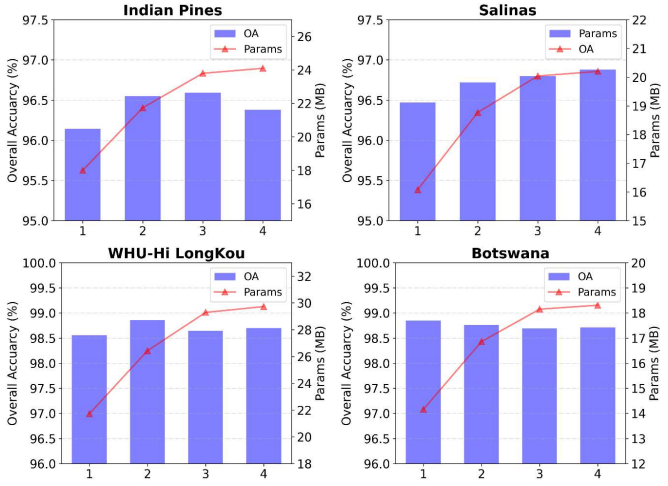


Fig. 12. Classification accuracy and memory occupation of Bi-SSFormer under different block numbers N, which is varied from 1 to 4. The OA (%) and Params (MB) on the four HSI datasets are reported.

4) Study on Patch Size and Partition Size: The patch size refers to the spatial size of the input for the entire HSI classification model, while the partition size P controls the window size and grid size for the self-attention operations in the SpaIM and needs to be divisible by the patch size. To explore their effect on the model performance and computational cost, we conduct a joint parameter analysis by evaluating different combinations of patch size and partition size, and we report the OA, AA, and FLOPs for the four datasets, as shown in Table X. From this table, we observe that the FLOPs increase with the patch size as this latter influences the spatial size of the internal feature maps. Notably, although the best classification performance is achieved with different patch sizes for various datasets, the optimal choice of partition size is invariably 2×2. This suggests that while our model benefits from global context provided by grid attention, a fine-grained local feature extraction (as enabled by smaller partitions) is consistently critical for achieving optimal performance across diverse datasets. Specifically, under this partition size, our model achieves the best OA of 96.80 $\pm$ 0.51% with a patch size of 10×10 for Indian Pines, 98.49 $\pm$ 0.46% with a patch size of 14×14 for Salinas, 98.86 $\pm$ 0.23% with a patch size of 8×8 for WHU-Hi LongKou, and 98.94 $\pm$ 0.35% with a patch size of 10×10 for Botswana. Considering that we aim to construct the most lightweight HSI model, we choose the patch size equal to 8×8 and partition size equal to 2×2 with the least computational costs as the default setting for all these datasets. In other scenarios with sufficient hardware resources, we can carefully choose the optimal patch size to get the best classification performance.

5) Study on Number of SS-Block: To investigate the impact of the number of SS-Block N on the model complexity and performance, we report the memory occupation and OA of Bi-SSFormer on the four HSI datasets in Fig. 12, where the number of blocks (N) is varied from 1 to 4.

As demonstrated in this figure, the OA of Bi-SSFormer improves significantly when the block number increases from 1 to 2 on Indian Pines, Salinas, and WHU-Hi LongKou

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TABLE XI

CLASSIFICATION RESULTS, MODEL COMPLEXITY, AND TRAINING TIME ON THE INDIAN PINES DATASET. BEST RESULTS ARE IN BOLDFACE. MB AND M STAND FOR MEGABYTE AND MILLION, RESPECTIVELY

Class	2-D CNN 2016	3-D CNN 2018	SpectralFormer 2022	SSFTT 2022	MorphFormer 2023	CAEVT 2022	GSC-ViT 2024	SPTNet 2024	CSCANet 2025	MambaLG 2025	Ours –
1	53.17±22.24	6.34±5.69	24.63±12.02	74.15±18.71	71.71±25.21	63.41±26.98	71.95±20.36	63.90±21.23	32.68±4.25	77.07±20.96	**84.63±14.18**
2	81.49±3.71	57.98±5.44	71.14±3.73	89.05±3.63	92.35±2.89	90.75±4.00	93.76±2.95	89.32±2.48	88.76±1.28	**94.48±3.15**	93.81±3.27
3	87.62±4.62	52.58±5.89	68.49±4.08	92.03±3.66	95.39±2.62	93.84±2.57	96.92±1.58	93.56±4.06	91.85±1.57	96.64±2.31	**96.83±1.72**
4	83.19±7.52	24.79±5.86	69.11±9.54	91.60±5.28	**93.80±6.06**	92.58±3.91	91.60±3.97	90.66±6.01	60.51±3.61	92.54±7.36	93.24±4.95
5	90.57±4.21	56.05±12.15	82.94±3.61	92.71±3.68	93.68±2.97	92.62±2.79	94.46±2.79	91.22±3.27	**95.13±0.62**	93.20±3.05	94.16±2.82
6	97.15±1.08	96.30±0.88	94.52±2.63	98.36±2.09	98.43±1.54	97.56±1.91	99.24±0.55	99.00±0.59	98.48±0.31	98.58±0.80	**99.28±0.49**
7	57.60±20.25	0.40±1.20	37.60±28.74	87.20±13.72	62.40±31.15	72.80±28.67	84.40±17.84	74.80±19.92	**93.60±6.50**	84.80±14.95	81.60±19.53
8	96.67±2.65	97.00±2.20	95.84±3.93	99.60±0.47	99.00±0.81	98.86±1.46	99.67±0.49	99.51±0.66	95.35±2.14	99.88±0.19	**100.00±0.00**
9	20.56±12.44	1.11±3.33	18.33±17.92	56.67±24.32	41.67±26.21	51.67±22.37	65.56±24.70	52.22±31.05	41.11±12.22	55.56±13.83	**83.33±12.91**
10	84.01±4.36	50.77±3.05	74.00±4.37	89.06±3.07	92.21±1.95	88.11±2.92	**95.23±2.11**	91.65±3.08	89.68±1.43	95.23±2.46	94.70±3.51
11	89.89±2.54	85.15±1.55	85.87±2.65	93.98±1.90	96.49±0.98	95.60±1.77	97.57±1.23	95.11±1.80	92.38±1.19	97.09±0.91	**97.57±0.77**
12	76.10±7.21	26.72±6.25	50.30±5.65	89.78±4.50	90.36±4.22	93.90±2.13	94.72±1.78	88.56±3.80	79.40±3.45	**95.22±2.14**	94.76±2.55
13	99.08±0.59	97.73±3.21	98.70±1.63	99.35±0.58	99.84±0.25	98.43±2.95	99.41±1.61	99.08±2.07	97.93±1.30	99.68±0.43	**99.95±0.16**
14	95.94±1.90	93.06±3.46	93.40±2.40	96.94±1.39	98.54±0.81	97.82±2.02	98.45±1.37	95.79±2.14	98.27±0.43	98.43±1.19	**98.74±1.12**
15	85.27±6.20	59.83±6.45	75.62±7.31	92.31±4.14	94.15±3.07	90.58±3.39	93.78±6.42	91.90±3.59	78.38±3.91	94.47±4.32	**96.97±2.06**
16	96.43±2.77	90.60±3.63	96.19±3.23	94.76±6.52	92.74±5.52	96.07±4.23	94.05±4.12	94.17±6.57	**97.71±1.82**	91.67±5.13	95.36±5.17
OA(%)	88.32±1.40	70.25±1.86	80.14±1.71	93.12±1.20	95.01±0.90	93.91±0.73	96.29±0.57	93.42±1.01	90.79±0.32	96.22±0.80	**96.55±0.43**
AA(%)	80.92±2.56	56.03±1.84	71.04±2.92	89.85±1.61	88.30±2.66	88.41±2.58	91.92±2.66	88.15±2.33	83.20±0.77	91.53±1.66	**94.06±1.73**
$\kappa \times 100$	86.68±1.61	65.44±2.20	77.31±1.97	92.16±1.37	94.31±1.03	93.05±0.83	95.77±0.65	92.50±1.15	89.47±0.36	95.69±0.91	**96.06±0.49**
Params (MB)	6.84	1.05	1.38	3.73	0.79	1.44	2.25	0.69	1.05	0.56	**0.02**
FLOPs (M)	17.84	45.20	35.98	16.46	19.46	120.80	20.96	9.49	7.11	10.30	**4.91**
Training Time (s)	159.09	177.29	294.02	163.83	266.45	207.76	163.53	194.44	59.35	99.39	176.35

TABLE XII

CLASSIFICATION RESULTS, MODEL COMPLEXITY, AND TRAINING TIME ON THE SALINAS DATASET. BEST RESULTS ARE IN BOLDFACE. MB AND M STAND FOR MEGABYTE AND MILLION, RESPECTIVELY

Class	2-D CNN 2016	3-D CNN 2018	SpectralFormer 2022	SSFTT 2022	MorphFormer 2023	CAEVT 2022	GSC-ViT 2024	SPTNet 2024	CSCANet 2025	MambaLG 2025	Ours –
1	70.40±44.51	98.38±1.39	92.63±2.95	99.92±0.21	99.94±0.17	99.69±0.64	**99.98±0.04**	99.45±0.54	99.93±0.18	99.97±0.07	99.89±0.20
2	86.75±29.40	99.75±0.14	98.89±0.96	99.21±1.50	99.92±0.10	99.03±1.37	99.54±0.99	99.26±0.68	99.78±0.11	99.93±0.10	**99.95±0.09**
3	79.66±24.78	88.70±3.65	90.54±6.16	94.49±3.78	96.96±2.47	96.38±3.12	96.29±4.14	95.34±3.40	**99.83±0.77**	99.08±1.62	98.81±2.08
4	98.10±2.76	99.04±0.73	93.39±3.23	**99.49±0.52**	99.14±1.09	99.21±0.96	97.84±3.41	99.22±1.05	98.96±0.41	99.00±2.01	98.63±3.06
5	93.23±3.19	94.76±2.51	87.50±3.39	97.85±1.88	97.10±1.20	95.67±3.61	**98.73±1.04**	96.46±1.60	98.16±0.20	97.84±1.34	97.33±2.73
6	99.85±0.24	99.97±0.05	99.80±0.40	99.78±0.47	**100.00±0.01**	99.96±0.08	99.95±0.14	99.97±0.07	99.57±0.44	99.99±0.04	99.97±0.05
7	99.54±0.22	99.67±0.11	97.58±2.06	99.72±0.43	99.91±0.13	99.77±0.42	99.89±0.18	99.61±0.42	98.95±0.43	**99.97±0.05**	99.96±0.07
8	85.61±4.13	84.61±1.85	83.60±3.55	86.98±3.01	91.23±2.22	90.33±3.27	92.12±3.08	87.92±2.76	**94.17±1.47**	90.21±2.86	91.25±2.10
9	98.53±1.88	98.17±1.11	96.79±1.90	99.20±0.72	99.75±0.39	99.36±0.76	**99.99±0.02**	99.04±1.22	99.98±0.03	99.99±0.02	99.91±0.27
10	86.70±13.37	90.78±2.73	89.22±3.53	95.52±2.10	95.66±1.99	94.79±2.11	96.88±1.00	95.72±1.80	**98.45±0.31**	96.49±1.97	97.02±1.39
11	49.58±39.87	69.84±12.88	78.35±5.78	94.55±5.00	97.08±2.42	92.36±6.71	**98.40±1.21**	93.82±4.62	96.36±1.08	97.42±2.63	97.41±2.88
12	94.79±4.86	98.47±0.90	96.67±1.54	99.70±0.31	99.84±0.31	98.84±2.30	99.31±0.76	97.99±3.64	**99.99±0.02**	98.53±1.80	98.69±2.06
13	89.65±29.89	**99.99±0.03**	97.93±2.35	99.69±0.44	99.31±0.86	99.23±1.03	99.79±0.43	99.49±0.71	64.23±18.69	99.87±0.27	99.65±0.69
14	81.83±31.88	**99.29±0.34**	98.68±0.86	97.16±2.36	98.45±2.29	98.51±1.47	98.06±2.20	97.01±1.98	96.59±1.71	99.14±1.10	97.97±2.57
15	73.69±17.20	75.27±2.55	76.59±5.14	85.25±2.38	90.11±3.74	86.37±6.10	90.99±4.10	86.16±4.37	84.45±2.89	90.55±4.43	**93.59±3.11**
16	75.58±19.11	91.08±4.40	91.75±6.49	96.63±2.91	97.36±3.15	96.72±2.78	98.24±1.57	97.03±2.69	98.27±1.28	97.57±3.08	**98.67±1.13**
OA(%)	86.79±8.03	90.95±0.72	89.88±1.06	94.24±0.45	96.07±0.24	94.96±0.69	96.51±0.45	94.43±0.46	95.56±0.54	96.10±0.48	**96.72±0.64**
AA(%)	85.22±11.87	92.99±0.87	91.87±1.31	96.57±0.45	97.61±0.36	96.64±0.68	97.88±0.43	96.47±0.61	95.48±1.31	97.85±0.42	**98.04±0.50**
$\kappa \times 100$	85.23±9.05	89.92±0.80	88.74±1.18	93.59±0.50	95.63±0.27	94.39±0.77	96.11±0.50	93.80±0.51	95.04±0.60	95.66±0.53	**96.35±0.71**
Params (MB)	6.87	1.05	1.42	3.80	0.81	1.44	0.41	0.70	1.05	0.56	**0.01**
FLOPs (M)	18.17	46.08	37.01	16.79	19.84	123.01	6.62	9.64	7.11	10.32	**4.15**
Training Time (s)	166.95	176.13	282.45	173.15	274.93	207.74	167.56	175.53	44.41	110.56	183.13

datasets. However, the accuracy increases marginally or even decreases when the number of blocks is further increased from 2 to 4. In contrast, the best classification accuracy is obtained when $N = 1$ on the Botswana dataset, which is slightly higher than that when $N = 2$. This phenomenon may be attributed to the regularization effect of a simpler model. In addition, for all datasets, memory requirements grow drastically with an increasing number of blocks, which poses challenges for the design of lightweight models.

In conclusion, to balance the model complexity and representational power for Bi-SSFormer, we opt for $N = 2$ as the default setting to ensure consistency across various datasets.

C. Comparison With SOTA Methods

To validate the effectiveness of the proposed super-lightweight model, we compare the classification results on the four HSI datasets with other SOTA deep learning-based methods, including 2-D CNN [34], 3-D CNN [35], SpectralFormer [36], SSFTT [43], MorphFormer [45], CAEVT [38], GSC-ViT [39], SPTNet [46], CSCANet [40], and MambaLG [52]. The model performance, complexity, and training time are provided in Tables XI–XIV. As shown in the results, the training time of Bi-SSFormer is acceptable (~2.5 min) and comparable with most of the SOTA works. We use the same default patch size of 8×8 for all compared SOTA methods to ensure a fair and direct comparison.

1) Numerical Results: For Indian Pines dataset, as shown in Table XI, the proposed model achieves the best OA with $96.55 \pm 0.43\%$ over other SOTA methods using only a few samples for training. Specifically, compared with the second-best model, GSC-ViT, our method gets 0.26% higher mean OA with 0.14% lower standard deviation, showing a more stable and superior classification performance in multiple running tests. In addition, the proposed model Bi-SSFormer obtains the best classification results on 9 out of 16 land cover categories, showing its remarkable generalization capability over different classes especially on classes with extremely limited training samples. For instance, Bi-SSFormer achieves an accuracy of

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TABLE XIII

CLASSIFICATION RESULTS, MODEL COMPLEXITY, AND TRAINING TIME ON THE WHU-HI LONGKOU DATASET. BEST RESULTS ARE IN BOLDFACE. MB AND M STAND FOR MEGABYTE AND MILLION, RESPECTIVELY

Class	2-D CNN 2016	3-D CNN 2018	SpectralFormer 2022	SSFTT 2022	MorphFormer 2023	CAEVT 2022	GSC-ViT 2024	SPTNet 2024	CSCANet 2025	MambaLG 2025	Ours –
1	96.48±2.19	98.65±0.50	99.44±0.34	99.15±0.71	99.09±0.69	99.37±0.75	99.52±0.36	99.13±0.56	99.91±0.04	99.67±0.31	**99.78±0.17**
2	59.22±31.13	80.41±9.64	76.97±10.76	97.28±2.26	96.74±2.13	93.16±4.15	96.67±1.92	91.46±3.79	96.46±1.23	98.19±1.63	**98.25±1.28**
3	4.04±12.12	69.37±11.00	88.78±3.72	91.53±6.70	89.64±7.40	88.71±6.72	91.47±4.49	85.24±6.49	91.99±1.83	**93.03±3.93**	90.55±4.47
4	94.42±2.27	96.85±2.15	96.80±1.30	98.50±0.48	99.07±0.47	98.26±1.19	99.20±0.52	98.69±0.92	**99.39±0.18**	99.02±0.54	99.29±0.36
5	9.83±16.28	25.21±12.28	53.10±22.02	80.73±15.46	83.99±8.55	85.57±12.61	86.06±9.74	77.09±8.32	66.04±4.76	88.40±4.31	**92.66±3.64**
6	77.52±29.37	97.45±2.38	94.08±2.10	97.52±1.83	98.04±1.34	98.71±1.29	98.55±2.10	97.54±2.42	**99.81±0.10**	98.84±1.09	99.25±0.93
7	99.99±0.03	99.96±0.04	99.86±0.18	99.89±0.13	99.93±0.09	99.95±0.05	99.90±0.11	**99.99±0.01**	99.97±0.01	99.91±0.10	99.89±0.11
8	50.55±34.40	89.10±4.16	82.08±6.11	91.99±2.52	91.16±2.19	93.42±2.85	91.36±3.21	85.90±5.35	**94.95±1.25**	92.92±1.92	93.35±1.94
9	22.24±17.47	66.99±10.24	42.93±7.25	89.43±2.38	84.01±8.48	87.64±5.54	86.53±5.53	79.55±4.94	**93.57±1.43**	92.36±3.03	91.82±3.46
OA(%)	87.74±4.47	94.64±0.76	94.38±1.15	98.04±0.33	98.09±0.34	97.98±0.43	98.37±0.21	97.25±0.38	98.48±0.17	98.69±0.16	**98.86±0.23**
AA(%)	57.14±10.69	80.44±1.90	81.56±3.91	94.00±1.91	93.52±1.52	93.86±1.76	94.36±1.17	90.51±1.11	93.56±0.85	95.82±0.89	**96.09±0.68**
$\kappa \times 100$	83.59±6.22	92.94±0.98	92.60±1.51	97.42±0.44	97.49±0.45	97.34±0.57	97.85±0.28	96.38±0.50	98.00±0.22	98.28±0.21	**98.50±0.30**
Params (MB)	7.48	0.81	2.17	5.02	1.02	1.82	0.69	0.84	1.04	0.57	**0.02**
FLOPs (M)	23.64	60.90	55.80	22.25	26.09	166.40	10.25	12.11	6.88	10.56	**5.82**
Training Time (s)	169.49	181.28	323.14	211.98	256.53	201.74	154.53	162.18	58.56	90.19	175.90

TABLE XIV

CLASSIFICATION RESULTS, MODEL COMPLEXITY, AND TRAINING TIME ON THE BOTSWANA DATASET. BEST RESULTS ARE IN BOLDFACE. MB AND M STAND FOR MEGABYTE AND MILLION, RESPECTIVELY

Class	2-D CNN 2016	3-D CNN 2018	SpectralFormer 2022	SSFTT 2022	MorphFormer 2023	CAEVT 2022	GSC-ViT 2024	SPTNet 2024	CSCANet 2025	MambaLG 2025	Ours –
1	99.96±0.12	**100.00±0.00**	99.79±0.28	99.51±0.82	99.79±0.28	99.88±0.37	97.98±2.94	99.63±0.87	99.70±0.60	98.97±0.91	98.23±2.30
2	82.31±10.33	87.91±9.87	86.26±13.59	93.74±14.24	98.46±2.47	95.93±5.16	98.79±3.28	97.80±3.61	85.00±7.40	99.45±1.13	**100.00±0.00**
3	91.37±3.82	97.35±1.31	91.28±4.11	98.05±2.90	99.38±0.97	99.56±0.66	99.47±0.97	78.72±39.39	99.51±1.48	100.00±0.00	**100.00±0.00**
4	79.85±8.92	95.31±3.54	70.72±11.60	95.57±6.55	**99.95±0.15**	99.07±1.51	98.92±1.95	80.41±33.12	99.27±1.47	99.69±0.93	99.43±0.75
5	78.06±8.55	81.49±6.91	35.17±8.39	92.64±6.77	92.52±6.88	95.54±4.36	90.83±9.00	77.81±27.34	89.96±2.63	95.21±4.41	**95.54±4.29**
6	81.24±8.84	77.48±6.50	71.20±12.84	92.36±5.39	96.45±3.47	96.12±3.16	96.40±3.62	80.95±30.05	91.21±3.52	97.19±2.06	**97.23±2.16**
7	98.24±1.28	77.48±6.50	98.11±1.30	99.83±0.28	99.01±1.37	99.79±0.52	99.87±0.39	89.79±29.93	**100.00±0.00**	97.85±6.44	99.96±0.13
8	77.87±6.98	83.50±10.14	69.62±10.42	97.81±3.53	98.58±3.02	98.74±2.03	98.74±2.58	66.99±35.70	98.41±2.67	99.95±0.16	**100.00±0.00**
9	76.47±6.16	95.09±4.98	57.42±14.54	95.16±5.84	97.46±3.07	98.41±2.51	98.76±1.20	79.26±32.77	**99.60±0.50**	97.99±3.02	98.27±2.73
10	93.90±5.51	95.78±1.89	91.79±7.26	96.28±5.52	99.10±0.94	98.97±1.39	98.21±2.63	82.42±31.64	98.08±4.15	99.01±1.98	**99.10±1.00**
11	97.31±3.22	98.65±1.41	97.89±1.91	94.84±10.56	99.42±0.77	97.53±3.70	98.95±1.32	78.76±27.32	**99.70±0.40**	96.47±6.23	99.24±1.03
12	96.63±4.06	99.20±0.83	94.48±2.88	99.08±1.32	99.20±0.91	98.83±2.34	**99.94±0.18**	81.47±32.44	98.31±1.88	99.45±1.46	99.75±0.30
13	94.94±5.16	97.80±2.63	93.57±2.90	98.84±2.23	99.09±1.52	99.67±1.00	99.67±0.58	79.67±39.63	97.91±3.76	**99.92±0.25**	99.79±0.43
14	93.41±3.76	80.47±27.01	82.59±2.28	95.76±6.78	96.12±5.49	**96.71±6.13**	95.29±5.62	93.53±10.11	87.56±7.73	95.06±4.90	95.88±3.25
OA(%)	88.76±1.45	92.86±1.33	80.85±1.70	96.42±1.86	98.20±0.75	98.33±0.95	98.01±0.83	82.42±21.88	96.91±1.32	98.29±1.38	**98.76±0.30**
AA(%)	88.68±1.66	92.05±2.51	81.42±1.90	96.39±1.99	98.18±0.66	98.20±1.01	97.99±0.88	83.37±20.43	96.02±1.55	98.30±1.26	**98.74±0.38**
$\kappa \times 100$	87.81±1.57	92.26±1.45	79.26±1.84	96.12±2.01	98.05±0.81	98.19±1.03	97.85±0.90	81.07±23.39	96.65±1.43	98.15±1.49	**98.65±0.32**
Params (MB)	6.33	0.72	0.92	2.71	0.78	1.05	0.41	0.58	1.05	0.54	**0.01**
FLOPs (M)	13.28	32.79	23.05	11.91	16.77	83.51	6.55	7.44	7.04	10.07	**3.67**
Training Time (s)	149.88	149.78	198.53	185.62	196.12	172.43	138.20	143.38	38.08	36.55	156.98

83.33% on Oats classes with only one sample, outperforming other SOTA methods such as the Mamba-based model MambaLG by 27.77% and the lightweight Transformer-CNN model CSCANet by 42.22%. Similarly, the classification accuracy of Hay-windrowed attains 100.00% with only 5% samples for training. This strong few-shot learning capability demonstrates the robustness and data efficiency of Bi-SSFormer. We attribute this to two core design principles: 1) the InWB acts as a powerful regularizer, preventing the model from overfitting to the few available samples and 2) the specialized SpeIM and SpaIM modules effectively extract highly discriminative spectral and spatial features for accurate HSI classification.

Regarding the model complexity, our method achieves 35× and 1.45× memory and computation savings, respectively, compared with the second-most-efficient model, demonstrating that our method strikes the best balance between model performance and complexity. The intrinsic reason for this superior balance lies in our core contributions. First, the InWB constrains weights to 1-bit values for extreme memory savings and eliminates the floating-point multiplications. Second, SpeIM and SpaIM use efficient convolution and linear-complexity attention mechanisms to extract the spectral and spatial features efficiently. Crucially, the information-enhanced forward and backward propagation strategy in InWB mitigates the performance loss typically associated with binarization, allowing Bi-SSFormer to remain highly accurate despite its extreme compression.

For the Salinas dataset, as presented in Table XII, our model obtains the best OA with 96.72 ± 0.64% and the least memory and computational requirements. Specifically, compared with the second-best model MambaLG, our Bi-SSFormer achieves 0.19% higher AA and 0.69 larger kappa coefficient with 56× and 2.49× fewer Params and FLOPs, respectively, significantly reducing the model complexity for efficient HSI classification. This top-ranking OA highlights the robust and balanced performance of Bi-SSFormer across all 16 land cover classes. Specifically, our method achieves SOTA accuracy on class 2 (Brocoli_green_weeds_2), class 15 (Vinyard_untrained), and class 16 (Vinyard_vertical_trellis). For these large and spectrally consistent regions, the proposed SpeIM–SpaIMs effectively learn subtle class-specific features, leading to exceptional performance. On the remaining classes, Bi-SSFormer delivers consistently strong and competitive results comparable with other leading methods. This balanced performance across diverse land cover types, achieved despite the representational constraints of 1-bit weights, robustly validates the superiority of our model design. This indicates that the primary advantage of our method lies in achieving SOTA

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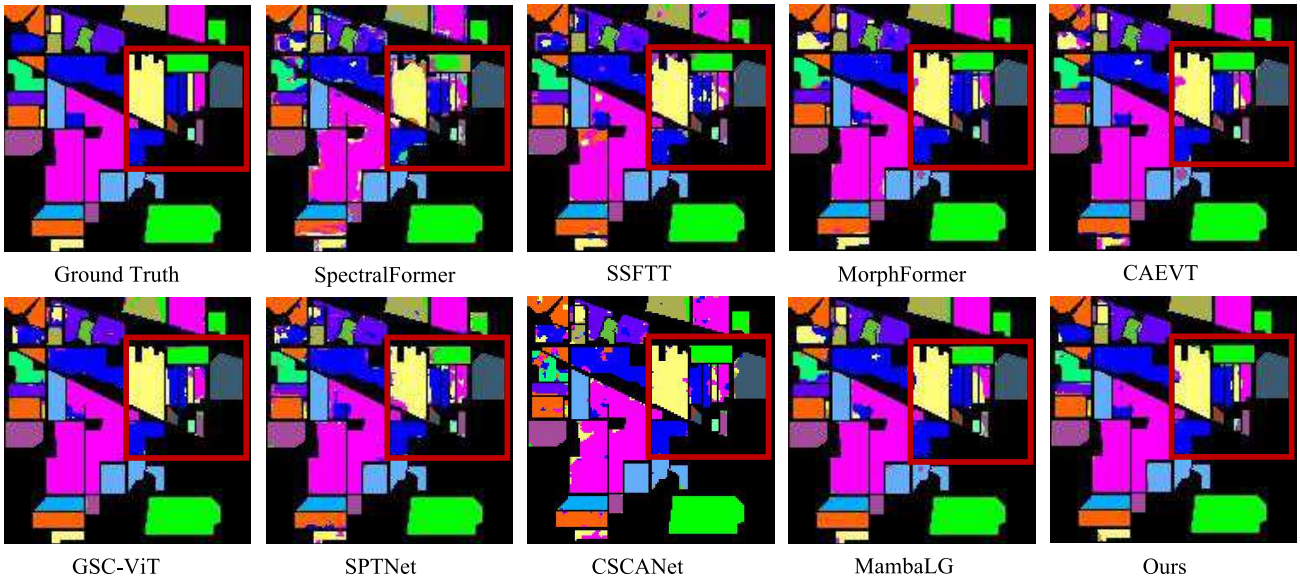


Fig. 13. Visualization of the classification map on the Indian Pines dataset, where different colors represent different categories according to Fig. 7.

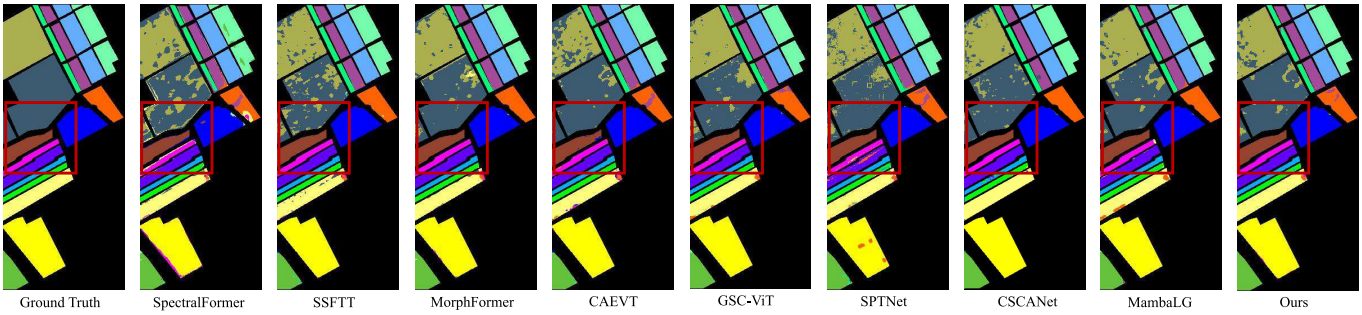


Fig. 14. Visualization of the classification map on the Salinas dataset, where different colors represent different categories according to Fig. 8.

accuracy while operating under scenarios where computational and memory resources are critically limited.

For the WHU-Hi LongKou dataset, Table XIII shows that our method achieves 0.17%, 0.38%, and 0.49% OA improvement over the SOTA approaches MambaLG, CSCANet, and GSC-ViT, respectively. Although the HSI classification performance is competitive with the advanced DL-based models, our super-lightweight architecture design and weight binarization technique enable the Bi-SSFormer to attain the most efficient inference budgets with only 0.02-MB memory requirements and 5.82M FLOPs for computation.

For the Botswana dataset, as reported in Table XIV, the proposed Bi-SSFormer still strikes the best tradeoff between classification performance and model complexity, obtaining $98.76 \pm 0.30\%$ OA and $98.74 \pm 0.38\%$ AA with only 0.01-MB parameter occupation and 3.67M FLOPs. It is worth to be remarked that our method achieves the best classification results on 50% of the land cover classes, especially for the Hippo-grass, Hippo grass-1, Hippo grass-2, and Island interior classes, whose accuracies are 100.00%.

2) Visual Results: To more intuitively show the advantages of the proposed model over other SOTA approaches, we visualize the classification maps of the four HSI datasets in Figs. 13–16. From these figures, we can observe that the proposed Bi-SSFormer provides more accurate and spatially consistent classification maps than the other SOTA methods.

For instance, in the classification map for Indian Pines dataset (see Fig. 13), our model generates smoother results with less noise, particularly within the challenging central region highlighted by the red box. Methods like 3-D CNN and SpectralFormer exhibit noticeable misclassifications and fragmented regions. In contrast, Bi-SSFormer produces larger and more homogeneous patches that are better aligned with the ground-truth map, especially in classes with only a few samples (Alfalfa). This visual superiority, especially at the boundaries between different land cover categories (Corn-notill, Soybean-notill, and Soybean-mintill), can be attributed to our SpaIM, which effectively integrates fine-grained local details and broader global context, preserving spatial details and reducing boundary artifacts. Similarly, for the Salinas dataset (see Fig. 14), our model shows cleaner edges and more accurate classification for large agricultural fields compared with most other SOTA methods, which tend to produce more noisy outputs. Finally, for the WHU-Hi LongKou and Botswana datasets (see Figs. 15 and 16), our proposed Bi-SSFormer still obtains smoother and more aligned classification maps than other SOTA models. Specifically, the narrow-leaf soybean and mixed weed classes in the WHU-Hi

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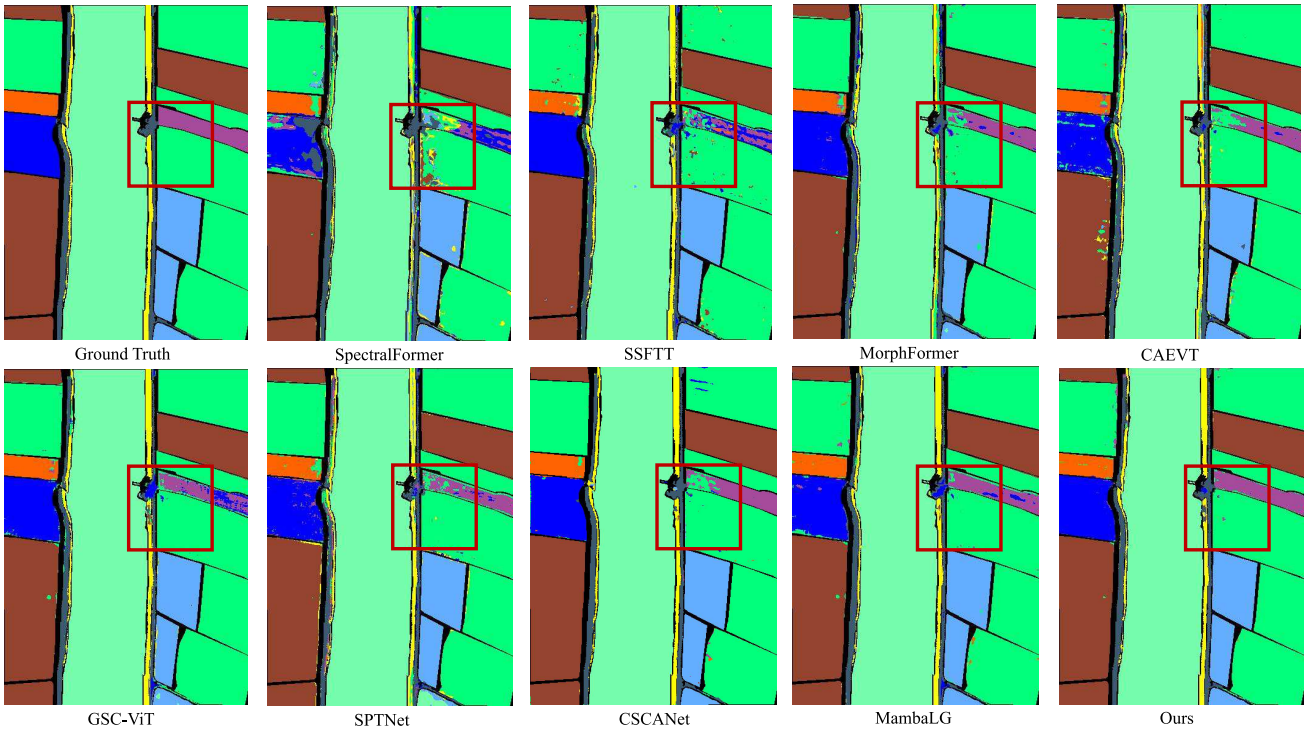


Fig. 15. Visualization of the classification map on the WHU-Hi LongKou dataset, where different colors represent different categories according to Fig. 9.

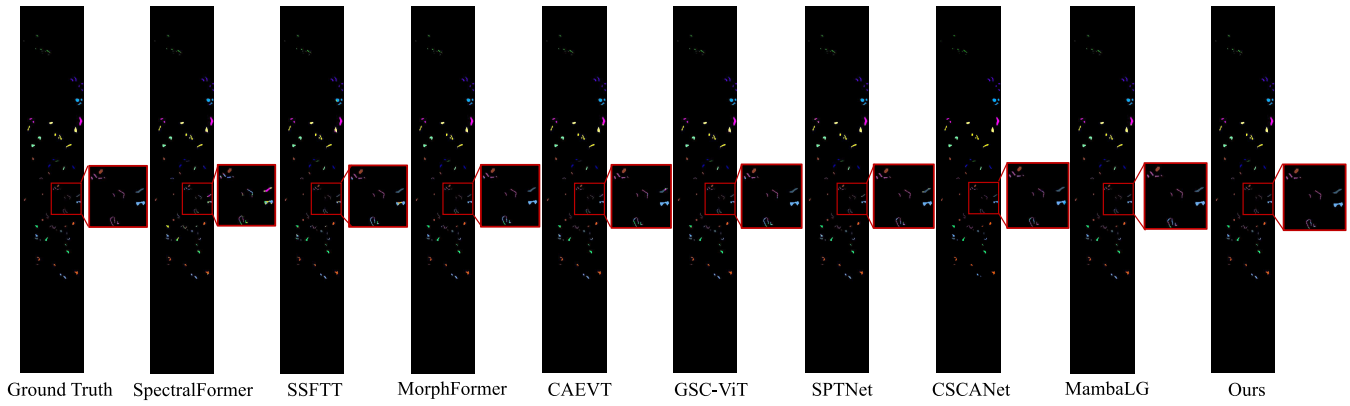


Fig. 16. Visualization of the classification map on the Botswana dataset, where different colors represent different categories according to Fig. 10. A closed-up is depicted in the red box.

LongKou dataset, denoted by the red box, demonstrate more accurate classification results with less noisy boundaries.

These results visually confirm the effectiveness of our model in leveraging both spectral and spatial information to achieve a more accurate and robust classification.

D. Analysis on Model Efficiency

While theoretical metrics such as FLOPs and Params provide initial estimates of model complexity, a comprehensive evaluation of a model's lightweight nature necessitates empirical testing on real-world hardware. To this end, we conduct a practical efficiency analysis to measure the actual inference speed and memory footprint of our proposed Bi-SSFormer on a single PC with an NVIDIA RTX 3090 GPU. We benchmark Bi-SSFormer against several competitive models, including GSC-ViT, SPTNet, and MambaLG, to provide a comparative assessment. It is important to note that CSCANet is excluded from this comparison due to its reliance on a PCA preprocessing step, which is unfair for evaluating the intrinsic efficiency of the models. Moreover, we leverage the highly optimized kernel implemented by BitNet [61] to accelerate the linear layers with binarized weights.

For the experiment, we evaluate the models on the Indian Pines dataset with a batch size of 64 and a patch size of 8. We measure the average inference time (ms) across ten runs and the peak GPU memory usage (MB) for efficiency analysis. The experimental results are summarized in Table XV.

As detailed in the table, our proposed Bi-SSFormer demonstrates superior practical efficiency. It achieves the lowest peak memory consumption and the fastest inference speed among

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TABLE XV

COMPARISONS OF ACTUAL MEMORY USAGE (MB) AND INFERENCE TIME (MS) ON THE INDIAN PINES DATASET. THE PATCH SIZE AND BATCH SIZE OF THE INPUT ARE SET AS 8 AND 64, RESPECTIVELY

Indian Pines	GSC-ViT	SPTNet	MambaLG	Ours
Memory Usage (MB)	61.86	60.94	61.93	**24.64**
Inference Time (ms)	189.99	117.56	164.49	**44.98**

all tested models. Specifically, Bi-SSFormer reduces memory usage by up to 2.51× and accelerates inference by up to 4.22× compared with other lightweight architectures. These empirical results robustly validate the lightweight properties of our model and underscore its suitability for efficient HSI classification.

Nevertheless, our current implementation only incorporates optimized kernels for linear computations. The development of custom, highly optimized kernels for binarized convolutions is a nontrivial task that we have not yet undertaken. Consequently, significant potential for further computational acceleration and memory savings in the convolutional layers remains. We leave the exploration and implementation of these optimizations as a promising direction for future work.

V. CONCLUSION

In this article, we propose a novel ultralightweight binary spectral–spatial transformer (Bi-SSFormer) to advance efficient HSI classification. The Bi-SSFormer integrates an SpeIM, an SpaIM, and an InWB technique. We first constructed a lightweight CNN–Transformer hybrid framework based on the SS-Block, where SpeIM and SpaIM extract representative spectral and spatial features, respectively. The SpeIM employs intragroup and cross-group convolutional layers to facilitate spectral feature exchange. The SpaIM leverages window attention and grid attention to capture local and global spatial information and then implements a complementary fusion scheme with feature enhancement for local–global interaction. Subsequently, the InWB technique binarizes all model weights to 1-bit representations, eliminating resource-intensive floating-point multiplications while substantially reducing memory requirements. Experimental results demonstrated that the proposed Bi-SSFormer achieves an optimal performance-complexity tradeoff, getting SOTA classification accuracy with minimal memory and computational overhead. Future developments go toward the study of advanced network pruning and knowledge distillation techniques, which can be considered in the proposed lightweight HSI classification network for further model compression.

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